



Truck scheduling with fixed outbound departures in a closed-loop conveyor system with shortcuts

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Abstract

With the global trend of e-commerce, companies pursue a higher quality of parcel delivery service since customers expect faster transportation in this fast-paced society. Several of them are committed to improving the efficiency of logistics and reducing operating costs and increasing their market competitiveness in the industry. The parcel distribution center thus plays a critical role in parcel delivery industries to sort and consolidate parcel flows to full truckloads. The benefit of the strategy is significant such as reducing transfer time and related costs. On the other hand, an automated sorting system (ASS) is also highly used in many supply chains with impressive characteristics like fast operation speed, large capacity, high reliability. The goal of this research is to apply the scheduling method to reduce the sorting time in an automated sorting system to improve distribution efficiency. This study focuses on the truck scheduling problem with fixed outbound schedules in a closed-loop conveyor system with shortcuts. The objective is to minimize the costs of extra trucks used to deliver delayed parcels and holding cost of parcels at each shipping dock door. If a parcel fails to be loaded onto the pre-determined outbound trucks, an extra outbound truck will be used to deliver the parcel. The problem is modeled with a mixed integer nonlinear programming model. This problem is proven to be NP-hard in the strong sense. As a result, an adaptive genetic algorithm with local search (LSAGA) is developed to solve the problem under twelve scenarios and compared with other algorithms, and a full factorial design of experiment was conducted. The computational experiments show that four factors, layout, inbound truck, outbound truck, and algorithm are significant to the objective value, and the proposed algorithm can obtain high-quality solutions with more stability. A sensitivity analysis is also conducted and bring some managerial insights.

Keywords Truck scheduling problem · Fixed departure · Closed-loop sorting system · Automated sorting system · Adaptive genetic algorithm

1 Introduction

With the vigorous development of e-commerce, the parcel service industry is a critical component of modern society. In today's fast-paced world, fast delivery of products can increase customer satisfaction and build companies' competitive advantage. Therefore, a large number of companies is highly focusing their efforts on improving efficiency in logistics operations to increase the reliability of delivery networks and provide higher service quality.

A parcel distribution center is an intermediate node in a distribution network serving as a hub of hub-and-spoke configuration. The consolidation of parcel flows in central hub terminals enables economies of scales of scale in transportation by unifying multiple smaller shipments to full truckloads (Boysen et al. 2017). As a result, the operation of a distribution center plays an important role in the distribution network of the parcel service industry.

An automated sorting system (ASS) often consists of receiving dock doors for receiving, a network system of conveyor, and shipping dock doors for shipping sorted parcels. With the advantage of fast, reliable, and large capacity, many supply chains apply an ASS as one essential component in their distribution concept. In the postal service industry, sorting mail and parcels is barely thinkable without ASS (Boysen et al. 2019). Furthermore, conveyor systems are widely used by parcel distribution center to reduce sorting cost and shorten transshipment time. In these parcel distribution centers, the inbound trucks are docked at receiving dock door, and parcels are unloaded from inbound trucks. After being unloaded, parcels will be scanned and verified, and then placed onto the conveyor systems. Parcels are moved to shipping dock doors through the sortation system and loaded into outbound trucks. Then, the outbound trucks leave the distribution center.

Truck scheduling problem is one important decision problem for ASS since the assignments of trucks may greatly affect the efficiency of sorting system. For instance, inbound trucks can be docked at different dock doors, and then the transshipment cost of shipments may be different. McWilliams et al. (2005) presented an inbound truck scheduling problem for parcel delivery industry with closed-loop conveyor systems. Several algorithms are proposed to solve truck scheduling problem with closed-loop conveyor systems (McWilliams 2009a, 2009b, 2010; McWilliams and McBride 2012; McWilliams et al. 2005). Chen et al. (2019) first considered a truck scheduling problem in a closed-loop conveyor system with shortcuts to increase throughput and improve efficiency. Alternative parcel routes from receiving to shipping dock must be considered with the existence of shortcuts. Moreover, the time of sorting process is highly affected by the travel time and traffic congestion of different parcel routes.

On the other hand, industries like postal services and less-than-truckload logistics transport relatively smaller and low value shipments. Delays of shipments are undesired but not as harmful as those of high valued shipments. As a result, these service providers typically rely on fixed outbound schedules (Boysen and Flidner 2010). An outbound truck must leave the hub before a certain moment no matter the assigned parcels arrive or not under this scenario. However, most previous

works studied truck scheduling problem with fixed outbound schedule where the internal transportation is executed manually (e.g. by workers using forklifts), but, as we discussed, ASS is widely applied in parcel delivery industries. Therefore, the truck scheduling in closed-loop conveyor with shortcuts under fixed outbound schedules deserve further study (see Fig. 1).

This research focuses on a truck scheduling problem for a closed-loop conveyor with shortcuts with fixed outbound schedules. Thus, this truck scheduling problem only schedules inbound trucks to receiving dock doors. According to fixed outbound schedules, we aim to schedule inbound trucks in the closed-loop system, so that the delivery cost of delayed shipments is minimized. The delivery cost comes from the extra outbound trucks used to deliver parcels that cannot be loaded on the pre-assigned outbound trucks. Also, this holding cost of parcels in temporary storage at shipping dock door is also considered. This problem is modeled in a mixed integer nonlinear programming model to minimize penalty delivery costs and holding cost.

The remainder of this research is organized as follows: in Sect. 2, we present a literature review on truck scheduling with related characteristics. Moreover, some previous studies about inbound truck scheduling with conveyor are introduced. In

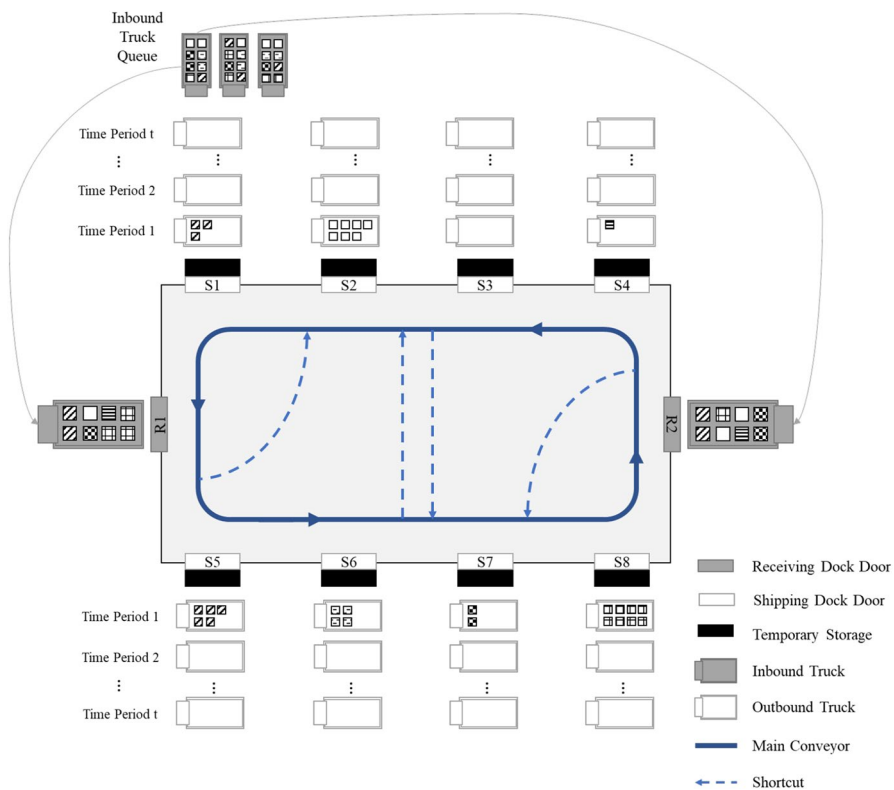


Fig. 1 Schematic layout under this scenario

Sect. 3, we describe the problem and develop a mathematical model. In Sect. 4, we introduce the methodology used to solve the proposed model. In Sect. 5, the computational results are reported. Finally, we provide the conclusions drawn from this research and propose perspectives.

2 Literature review

Cross-docking is known of a logistic strategy and used in distribution networks. Boysen and Flidner (2010) reviewed literature about cross-dock scheduling and provided a classification of cross-dock scheduling problem based on a tuple notation. Several papers extended the classification provided by Boysen and Flidner (2010) to categorize articles of cross-docking (Ladier and Alpan 2016; Van Belle et al. 2012). Van Belle et al. (2012) published a state-of-art of cross-docking, which presented the overview of the cross-docking concept and discussed characteristics and implementations of cross-docking. Boysen et al. (2019) proposed a survey of automated sortation conveyors (ASS) and described ASS in warehouses, cross docks, postal service industry, and airports. The authors also provided some characteristics of ASS and defined decision problems to be solved.

2.1 Automated sortation system in distribution industry

McWilliams et al. (2005) defined the parcel hub scheduling problem (PHSP), considering inbound truck scheduling problem with material handling system in the parcel delivery industry. After unloading, parcels are transshipped to outbound trucks with a network of conveyors. The transshipment time is then dependent on its route and congestion of the conveyor network. The objective of PHSP is mostly to minimize the time span of the transfer operation involving unloading parcels from inbound trucks, sorting, and loading onto outbound trucks. Moreover, a simulation-based scheduling algorithm (SBSA) with a genetic algorithm (GA) was used to solve the PHSP by McWilliams et al. (2005).

The PHSP was modeled as a minimax problem where the objective was to minimize the maximum workload on the parallel output stations in the work of McWilliams (2009b) and McWilliams (2010). A decomposition method with a combination of time-based and resource-based was proposed to solve this problem (McWilliams 2009b, 2010). McWilliams (2009b) proposed a GA to solve large-size problems with less computational time compared to SBSA. McWilliams (2010) released the assumption of same batch size of inbound trucks and solved the same problem by applying a local search algorithm and simulation annealing.

McWilliams (2009a) presented a dynamic load-balancing algorithm (DLBA) to solve the PHSP. The objective function was to minimize the penalty for unbalanced workload. The performance of DLBA was compared with that of an efficient static load-balancing algorithm (SLBA). The solutions offered by DLBA are 8.6% superior to those offered by SLBA for large-size problems. McWilliams and McBride (2012) proposed a beam search scheduling heuristic to solve the PHSP. The performance

of the beam search heuristic is better than that of GA, simulation-based scheduling, and random scheduling. Chen et al. (2019) considered adding shortcuts into ASS to improve the speed and efficiency of parcel sorting flows. AGA was applied to minimize make span, and fuzzy logic control is used to adjust the probability of crossover and mutation rates.

Boysen et al. (2017) investigated the truck scheduling problem in the postal service industry. The impact of truck scheduling on the sortation performance of the central conveyor system were also presented. The fully automated sortation conveyor was used to connect inbound and outbound dock segments. Decomposition procedures were proposed for truck scheduling problem in the postal service industry (TSPS).

2.2 Cross-dock scheduling with destination mode of product allocation

Van Belle et al. (2012) classified product interchangeability into two types, pre-distribution, and post-distribution. The third type of product interchangeability, destination, was proposed by Ladier and Alpan (2016). Serrano et al. (2017) redefined this characteristic as product allocation and isolated the characteristic of interchangeability which indicates that the products are replaceable or not. In a destination mode of product allocation, the destinations of all shipments are known. All shipments are dedicated to a specific destination and outbound truck in a pre-distribution mode. In a post-distribution, in addition to destination of shipments, the types of shipments and the corresponding quantity to be loaded on outbound trucks are known for each outbound truck. In this part, we reviewed papers about cross-dock scheduling with destination mode. Several works of literature dealt with cross dock scheduling under product allocation with destination mode.

Serrano et al. (2017) proposed a MILP to minimize penalty costs associated with inbound truck arrival times and presented a case study of Renault ILN cross-dock. This model considered not only truck scheduling but also internal operations (including storing and repacking). Moreover, the utilization of the resource was also discussed in the model as the internal resources are limited.

Ladier and Alpan (2018) proposed to schedule inbound and outbound truck simultaneously under consideration of internal transformation. Inbound content was known, so the destination of pallets was available before the scheduling. Outbound trucks cannot leave before they are fully loaded, and the resource capacity was limited. The planning horizon was cut into a set of time periods. Since there was a preferred time window for arrival and departure of trucks, both the earliness and tardiness were considered as dissatisfaction. The goal was to minimize both the total number of pallets put in storage and the dissatisfaction of the transportation providers.

2.3 Truck scheduling with fixed outbound schedule

Boysen and Fliedner (2010) proposed a simplified model of cross ducking with fixed outbound schedules. It is identified that truck scheduling with fixed outbound

schedules should be considered in order to implement truck scheduling to real world. Several recent articles have been devoted to truck scheduling with fixed outbound truck departures.

Boysen et al. (2013) considered a truck scheduling problem with fixed outbound departures. The authors extended the model proposed by Boysen and Flidner (2010) to minimize total lost profit. Liao et al. (2013) considered simultaneous door assignment and sequencing of inbound trucks under fixed outbound departures. Six metaheuristic algorithms were presented for a multi-door cross docking operation to minimize total weight tardiness.

Tootkaleh et al. (2016) studied a scheduling problem of inbound trucks based on a fixed outbound truck schedule. Delayed loads were stored in temporary storage until the next outbound truck with the same destination is available. Each outbound door was assigned to a specific customer. Moreover, under the concept of post-destination, loads of other destinations can be replaced with delayed loads to reduce the total inventoried transferred to the next outbound departure time. A mixed integer linear model was developed to minimize total inventory holding cost.

Molavi et al. (2018) studied a simultaneous inbound and outbound truck scheduling problem with a fixed outbound truck departure schedule where the sequence of unloading shipments is also considered. The authors developed a mixed integer programming model to minimize the total cost comprising penalty and delivery cost of delayed shipments.

To best of our knowledge, we are the first to study truck scheduling with fixed departures in a closed-conveyor system with shortcuts. This paper refers to the consideration of Chen et al. (2019), and fixed outbound truck schedules is considered to make truck scheduling closer to reality. As mentioned above, several characteristics are proposed by Boysen et al. (2019); Boysen and Flidner (2010); Ladier and Alpan (2016); Van Belle et al. (2012). We adopt characteristics related to this research to summarize the literature review. Table 1 shows the summary of related literatures.

3 Problem definition and system description

This research considers an inbound truck scheduling problem with fixed outbound schedule in a cross-dock facility with a conveyor system. The objective is to minimize the costs of extra trucks used to deliver delayed parcels.

We consider a parcel distribution center with multiple receiving and shipping dock doors, and a set of outbound trucks is scheduled in advance. In other words, for each outbound truck, the departure time and destination are given. Moreover, each shipping dock door is assigned to a specific destination, and the dock door to be docked is also predetermined for outbound trucks. Consequently, outbound trucks with same destination are assigned to the same set of dock doors. It is assumed that the number of receiving door is fewer than or equal to the number of inbound trucks.

On the other hand, a conveyor system is used for internal transshipment, and it is equipped with shortcuts to reduce sorting time. When a parcel is unloaded from inbound truck, it is then transhipped to the corresponding shipping dock door of its destination through conveyor system as the destination information of each parcel is

Table 1 Summary of related literatures

Author	Problem	Doors inbound/outbound	Internal transformation	Service mode	Product allocation	Product interchangeability	Truck interchangeability	Pre-emption	Temporary storage Capacity	Handling capacity	Arrival time	Deadlines
McWilliams et al. (2005)	In	> 1/> 1	Net	Excl	Dest	No	No	No	Zero	∞	Zero	No
McWilliams (2009a)	In	> 1/> 1	Net	Excl	Dest	No	No	No	Zero	Limited	Zero	No
McWilliams (2009b)	In	> 1/> 1	Net	Excl	Dest	No	No	No	Zero	Limited	Zero	No
McWilliams (2010)	In	> 1/> 1	Net	Excl	Dest	No	No	No	Zero	Limited	Zero	No
Alpan et al. (2011)	In, out	> 1/> 1	NEM	Excl	Dest	No	No	Yes	∞	∞	Pred	No
McWilliams and McBride (2012)	In	> 1/> 1	Net	Excl	Dest	No	No	No	Zero	Limited	Zero	No
Boysen et al. (2013)	In	> 1/> 1	Manu	Excl	Pre-D	No	No	No	∞	∞	Zero	Out
Liao et al. (2013)	In	> 1/> 1	Manu	Excl	Post-D	Yes	No	No	∞	∞	Zero	Out
Tootkaleh et al. (2016)	In	> 1/> 1	Manu	Excl	Post-D	Yes	No	No	∞	∞	Zero	Out

Table 1 (continued)

Author	Problem	Doors inbound/outbound	Internal transformation	Service mode	Product allocation	Product interchangeability	Truck interchangeability	Pre-emption	Temporary storage Capacity	Handling capacity	Arrival time	Deadlines
Boysen et al. (2017)	In	> 1/> 1	loop	Excl	Pre-D	No	No	No	Zero	∞	Pred	Out
Serrano et al. (2017)	In, out	> 1/> 1	Manu	Excl	Dest	Yes	No	No	Limited	Limited	Pred	No
Ladier and Alpan (2018)	In, out	> 1/> 1	Manu	Excl	Dest	No	No	No	∞	Limited	Pred	In, Out
Molavi et al. (2018)	In	> 1/> 1	Manu	Excl	Pre-D	No	Yes	No	∞	∞	Pred	Out
Chen et al. (2019)	In	> 1/> 1	Net	Excl	Dest	No	No	No	zero	Limited	Zero	No
This paper	In	> 1/> 1	Net	Excl	Dest	No	No	No	∞	Limited	Zero	Out

Problem: *In* inbound problem, *Out* outbound problem. Internal transformation: *Manu.* manually, *Loop* loop sorter, *Net.* conveyor network, *NEM* not explicitly modeled. Service mode: *Excl.* exclusive. Product allocation: *Dest.* destination, *Pre-D* pre-distribution, *Post-D* post-distribution. Arrival time: *Pred.* predefined

identified. There might be routes with shortcut from the receiving dock door to shipping dock door, and, therefore choosing a route with or without shortcut might be a decision task. A parcel will be directly loaded onto the outbound truck if there is an outbound truck which is not fully loaded docked at the dock door when it arrives. Otherwise, it will be temporarily placed in the storage next to dock door.

All parcels arriving at the distribution center each day should leave the distribution center on the same day. With consideration of limited capacity and fixed departure schedule, the originally scheduled outbound trucks may not be able to deliver all parcels of the day. As a result, extra trucks will be used to deliver delayed parcels. As extra truck is also assigned to a specific destination, parcels with different destination need distinct extra trucks for delivery.

The remaining assumptions are as follows:

1. All receiving dock doors in the conveyor system are identical and can process any inbound truck.
2. All inbound trucks present equal priority.
3. Inbound trucks are all available at the beginning of scheduling horizon.
4. Inbound trucks are not allowed for pre-emption. Once docked, an inbound truck cannot leave the dock door until it is completely unloaded.
5. An outbound truck is docked at a shipping dock door for a certain period of time and departure at a pre-decided time.
6. Outbound trucks have a limited capacity. The capacity is identical to outbound trucks. Parcels cannot be loaded once the outbound truck is full, even if parcels arrive at the dock door before departure of trucks.
7. A conveyor system is equipped with main conveyor and shortcuts to transship parcels from receiving dock doors to shipping dock doors.
8. Each outbound truck is pre-assigned to a specific outbound door.
9. There is a temporary storage with unlimited capacity at each shipping dock door.
10. There is more than one door at both receiving and shipping docks, and the service mode of doors is exclusive.
11. Any parcel arriving at the terminal is dedicated to a specific destination. Parcels are not interchangeable.
12. The sequence of parcels unloaded from inbound trucks is randomly generated.

3.1 Notation and system description

3.1.1 Notation

The notations of mathematical model are defined as shown in Table 2.

3.1.2 Composition of routes

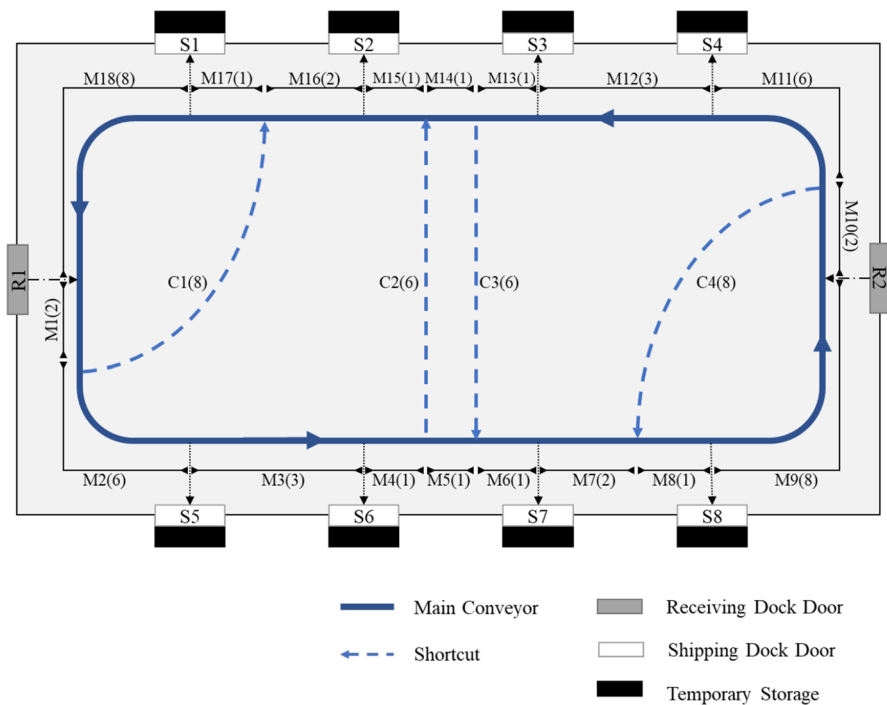
Figure 2 shows an example of conveyor system with shortcuts. The conveyor system contains a single closed-loop main conveyor (M1–M18), four shortcuts (C1–C4), two receiving dock doors (R1 and R2), and eight shipping dock doors (S1–S8).

Table 2 Notations of mathematical model

<i>Sets</i>	
I	Set of inbound trucks with $I = \{1, 2, \dots, n\}$
O	Set of outbound trucks
R	Set of receiving dock doors available for processing inbound trucks
S	Set of shipping dock doors available for processing outbound trucks
T	Set of time slots
M	Set of conveyor sections (including mainline conveyor and shortcuts)
K	Set of groups
K_i	Set of groups of parcels transported from inbound truck i
K_s	Set of groups of parcels whose destination is shipping dock door s
<i>Indices</i>	
i	Index value of inbound trucks
o	Index value of outbound trucks
r	Index value of receiving dock doors available for processing inbound trucks
s	Index value of shipping dock doors available for processing outbound trucks
t	Index value of time slots
m	Index value of conveyor sections
k	Index value of groups
<i>Parameters</i>	
p_i	Process time for unloading inbound truck i
d_o	Fixed departure time of outbound truck o
q_k	Shipping dock door of group k
w_k	Weight, e.g., the number of parcels in group k
c	Capacity limitation of outbound truck
tc	Increased cost per extra outbound truck for delivering delayed parcels
hc	Holding cost per group of parcels for one time slot
cm_d	Directly reachable conveyor section from dock door d (including shipping and receiving) $d \in G \cup S$
$BigM$	Big number
$e_{k,o}$	Binary parameter: 1, if outbound truck o is heading for the destination of group k ; 0, otherwise i.e., outbound truck o is docked at shipping dock door s , and the destination of group k is shipping dock door s ; 0, otherwise
g_{ij}	Binary parameter: 1, if conveyor segment j is directly reachable from conveyor segment i to; 0, otherwise
<i>Variables</i>	
B_k	Receiving dock door of group k
C_i	Completion time of inbound truck i
ST_k	Start time of group k
ET_k	End time of group k
ET_{max}	Maximum end time of all groups
N_s	Amount of extra outbound trucks used to transport delayed parcels at shipping dock door s
$Z_{k,m,t}$	Binary variable: 1, if group k is processed on conveyor segment m in time slot t ; 0, otherwise
$Z_{k,t}^{start}$	Binary variable: 1, if group k is processed first in time slot t ; 0, otherwise

Table 2 (continued)

$z_{k,t}^{end}$	Binary variable: 1, if group k is processed last in time slot t ; 0, otherwise
$H_{k,o}$	Binary variable: 1, if group k is loaded onto outbound truck o ; 0, otherwise
$X_{r,i,j}$	Binary variable: 1, if inbound truck j is processed directly after inbound truck i at receiving dock door r ; 0, otherwise
$X_{r,0,i}$	Binary variable: 1, if inbound truck i is processed first at receiving dock door g ; 0, otherwise
$X_{r,i,n+1}$	Binary variable: 1, if inbound truck i is processed last at receiving dock door r ; 0, otherwise

**Fig. 2** Conveyor system with shortcuts

The main conveyor is divided into sections with identical lengths. To explain more clearly, the sections between decision points, including receiving dock doors, shipping dock door, converging point, and diverging point, compose main conveyor segments, M1 to M18, and shortcut segments, C1–C4. For example, M18 comprise eight sections between S1 and R1.

The combination of segments can create multiple counterclockwise routes from receiving door to shipping door. Each route cannot contain duplicate segments, that is, parcels cannot pass through one segment twice. There are twenty-two possible routes in this conveyor system. Take R1 for example, there is only one possible

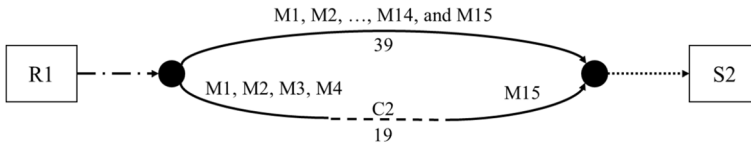


Fig. 3 Routes between R1 and S2

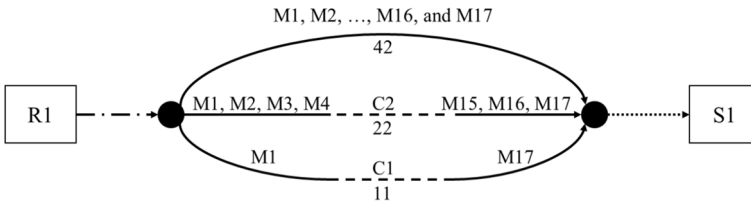


Fig. 4 Routes between R1 and S1

route which is totally composed of segments of main conveyor for S3–S8. Namely, every parcel unloaded at R1 and assigned to S7 can only be transhipped through the route comprises M1–M6. However, there exist two routes between R1 and S2 (see Fig. 3). In addition, three routes are available when a parcel is unloaded at R1 and assigned to S1 (see Fig. 4). The first route (P1) contains only main conveyor segments (M1–M17), and the total travel distance is forty-two. The second route (P2) is made up of segments from main conveyor (M1–M15) and shortcut (C2), and the total distance is twenty-two. The third route (P3) contains only three segments (M1, C1, and M17) where the travel distance is the shortest, eleven. It is obvious that routes with shortcuts are shorter than the one without shortcut, and thus parcels require less time to travel along P3 and P2. As a result, we prefer to transfer parcels via P3 rather than P1 and P2 to reduce transfer time.

3.1.3 Grouping for transshipment

Pallets are used to transship parcels from receiving dock doors to shipping dock doors. The capacity is identical to pallets. There will be no stopovers on the way from a receiving dock door to a shipping dock door. In other words, pallets are with single starting point and single destination, so parcels are classified by destination at receiving dock door and formed into groups in identical capacity to put on pallets. If parcels with same destination in an inbound truck is not divisible by the capacity of pallet, the remaining will form into a group and put on one pallet, and no other parcels will be put onto the pallet. Therefore, only parcels with same destination from the same inbound truck will be put on the same pallet. For example, if an inbound truck is composed of 38 parcels to destination A, 19 parcels to destination B, and 26 parcels to destination C, and the capacity of a pallet is 10. Then, parcels to destination A will form into 4 groups (10, 10, 10, 8). Parcels to destination B will form into 2 groups (10, 7), and parcels to destination C will form into 3 group (10, 10, 6). (see Fig. 5).

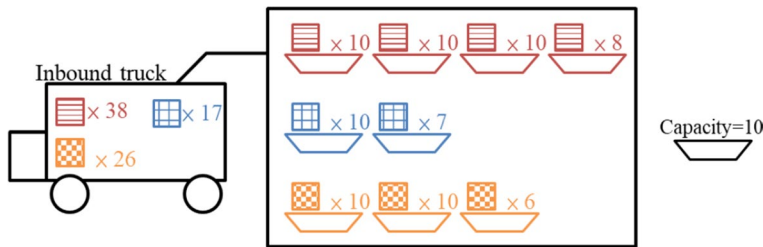


Fig. 5 Grouping principle of pallets

Table 3 Converging and diverging policy

Type	Policy
Converging point	1. The group on main conveyor has higher priority 2. The group on shortcut has higher priority 3. The group which is further from its destination has higher priority 4. The group with lower loading rate of its destination has higher priority 5. The group with fewer WIP of its destination has higher priority
Diverging point	1. The route with main conveyor has higher priority 2. The route with shortcut has higher priority 3. Randomly select one route

3.2 Converging and diverging policy

Since this paper considers a conveyor system with shortcut, the policy to decide the priority of sections adopted at each converging or diverging point may also be a critical factor for transshipment time. Consequently, several policies are proposed in this section. Table 3 shows converging and diverging policies proposed for this conveyor system.

However, if a converging point always let the group on specific section go first, it is possible that a group on another section may detain at the same section for a long period of time. As shown in Fig. 6, group $k1$ and $k2$ are about to converge into one section. If the policy gives the group on shortcut higher priority, $k2$ goes first at time t . The following $k3$ converge with $k1$ at time $t + 1$, and $k3$ still has higher priority as it is the one on shortcut. On condition that $k4$, $k5$, and $k6$ follow behind $k3$, $k1$ may detain at the same section for 4 periods of time. To avoid groups detained at one section, a mechanism is set that the group stay at the same section for two time periods has highest priority. Figure 7 shows the route planning for groups after applying the mechanism.

For the diverging points, it is worth mentioning that only the group with destination reachable from the shortcut can choose to go on shortcut. Otherwise, the group may go into infinite loop. An example is shown in Fig. 8. Assuming that the destination of $k1$ is $S3$, if the diverging policy of diverging point, $D1$, is the

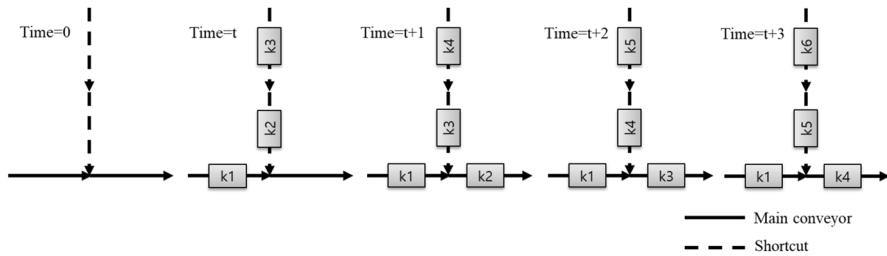


Fig. 6 Schematic diagram of detained group

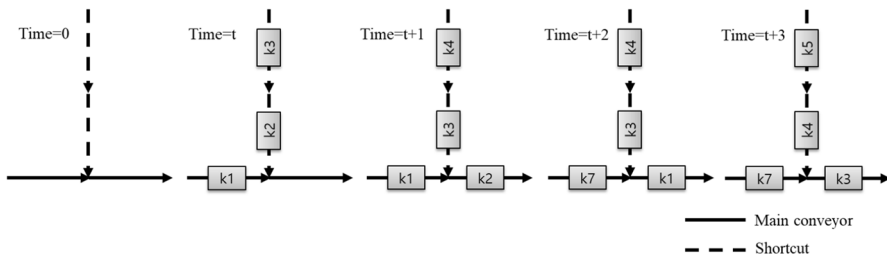


Fig. 7 Schematic diagram of the mechanism to prevent detention

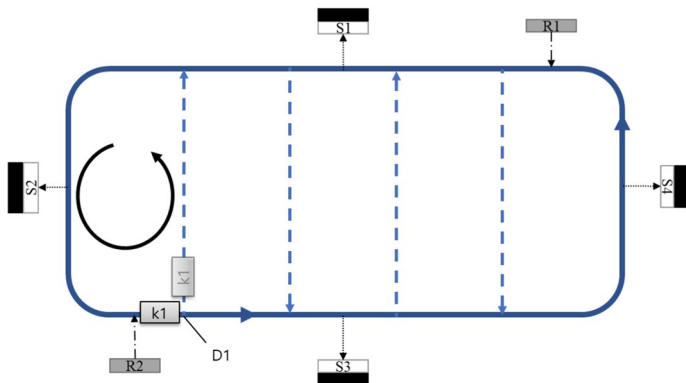


Fig. 8 Schematic diagram of groups that go into infinite loop

route with shortcut has higher priority, k_1 will choose to go route with shortcut which is unreachable to its destination and go into an infinite loop. Therefore, only the groups with reachable destination are allowed to go onto shortcuts.

3.3 Problem formulation

$$\text{Minimize } \sum_{s \in S} N_s \times tc + \sum_{k \in K} \left(\left(\sum_{o \in O} d_o \times H_{k,o} + \left(1 - \sum_{o \in O} H_{k,o} \right) \times ET_{\max} \right) - ET_k \right) \times hc.$$

Subject to

$$\sum_{r \in R} \sum_{\substack{j \in I \cup \{0\} \\ i \neq j}} X_{r,j,i} = 1 \quad \forall i \in I \quad (1)$$

$$\sum_{j \in I} X_{r,0,j} \leq 1 \quad \forall r \in R \quad (2)$$

$$\sum_{\substack{j \in I \cup \{0\} \\ i \neq j}} X_{r,j,i} = \sum_{\substack{j \in I \cup \{n+1\} \\ i \neq j}} X_{r,i,j} \quad \forall i \in I; \quad r \in R \quad (3)$$

$$C_i \geq C_j + p_i - \text{Big } M \times (1 - X_{r,j,i}) \quad \forall i \in I; \quad j \in I \cup \{0\}; \quad r \in R \quad (4)$$

$$C_0 = 0 \quad (5)$$

$$B_k = \sum_{r \in R} \sum_{\substack{j \in I \cup \{0\} \\ i \neq j}} X_{r,j,i} \times r \quad \forall i \in I; \quad k \in K_i \quad (6)$$

$$ST_k = \sum_{t \in T} Z_{k,t}^{\text{start}} \times t \quad \forall k \in K \quad (7)$$

$$ET_k = \sum_{t \in T} Z_{k,t}^{\text{end}} \times t \quad \forall k \in K \quad (8)$$

$$Z_{k,m,t} = 0 \quad \forall k \in K; \quad m \in M; \quad t \in T; \quad t < ST_k \quad (9)$$

$$Z_{k,m,t} = 0 \quad \forall k \in K; \quad m \in M; \quad t \in T; \quad t > ET_k \quad (10)$$

$$ST_k \geq C_i \quad \forall i \in I; \quad k \in K_i \quad (11)$$

$$Z_{k,m,t} = 1 \quad \forall k \in K; \quad m = cm_{B_k}; \quad t = ST_k \quad (12)$$

$$Z_{k,m,t} = 1 \quad \forall k \in K; \quad m = cm_{q_k}; \quad t = ET_k \quad (13)$$

$$\sum_{k \in K} Z_{k,m,t} \leq 1 \quad \forall m \in M; \quad t \in T \quad (14)$$

$$\sum_{k \in K} Z_{k,m,t} \leq 1 \quad \forall k \in K; \quad t \in T \quad (15)$$

$$\sum_{n \in M} g_{m,n} \times Z_{k,n,t+1} \geq Z_{k,m,t} \quad \forall k \in K; \quad t \in T; \quad m \in M - \{cm_{q_k}\} \quad (16)$$

$$Big M \times (1 - Z_{k,m,t+2}) \geq Z_{k,m,t+1} + Z_{k,m,t} - 1 \quad \forall k \in K; \quad t \in T; \quad m \in M - \{cm_{q_k}\} \quad (17)$$

$$Big M \times (2 - Z_{k,m,t} - Z_{k,m,t+1}) \geq \sum_{\substack{n \in M \\ n \neq m}} g_{m,n} \times \left(1 - \sum_{\substack{j \in K \\ j \neq k}} Z_{j,n,t+1}\right) \quad (18)$$

$$\forall k \in K; \quad t \in T; \quad m \in M - \{cm_{q_k}\}$$

$$\sum_{o \in O} H_{k,o} \leq 1 \quad \forall k \in K \quad (19)$$

$$\sum_{k \in K} w_k \times H_{k,o} \leq c \quad \forall o \in O \quad (20)$$

$$Big M \times [d_o - ET_k]^+ \geq H_{k,o} \quad \forall o \in O; \quad k \in K \quad (21)$$

$$e_{k,o} \geq H_{k,o} \quad \forall o \in O; \quad k \in K \quad (22)$$

$$\sum_{k \in K_s} w_k \times \left(1 - \sum_{o \in O} H_{k,o}\right) \leq c \times N_s \quad \forall s \in S \quad (23)$$

$$ET_{\max} \geq ET_k \quad \forall k \in K \quad (24)$$

$$Z_{k,m,t}, H_{k,o}, X_{r,0,i}, X_{r,i,j}, X_{r,i,n+1} \in \{0,1\} \quad \forall i \in I; \quad o \in O; \quad m \in M; \quad r \in R; \quad k \in K; \quad t \in T; \quad s \in S \quad (25)$$

$$B_k, C_k, ST_k, ET_k, ET_{\max} \geq 0 \quad \forall i \in I; \quad k \in K; \quad s \in S \quad (26)$$

$$N_s \in N^+ \cup \{0\} \quad (27)$$

The objective function is to minimize the extra truck cost used to transport delayed parcels as well as holding cost of parcels at shipping dock door. Constraints (1) ensure that each inbound truck is assigned to one receiving dock door. Constraints (2) make sure that no more than one sequence of inbound truck is assigned to each receiving dock door. Constraints guarantee the inbound truck sequence of dock door. Constraints define the completion time of each inbound truck considering the process time of unloading parcels to conveyor system. Constraints (5) set the completion time of virtual start inbound truck 0 to zero. This inbound truck scheduling model is based on Boysen et al. (2013). In constraints, the decision variable B_k is calculated. Once an inbound truck is assigned to receiving dock door r , then any group k on the truck are assigned to receiving dock door r . Constraints (7) and calculate the start time ST_k and end time ET_k of transshipment of group k . Constraints (9) and ensure groups are neither put on conveyor before ST_k nor after ET_k . Constraints (11) guarantee that groups are put onto conveyor system after the inbound truck is docked to receiving dock door. Constraints

(12) and ensure that for groups are put on section connected with receiving and shipping dock door for groups at ST_k and ET_k . Constraints ensure that there is no more than one group on section m at time t . Constraints make sure that each group stays at no more than one section at time t . Constraints guarantee that group move from current section to next connected section as time goes on. As $g_{m,m} = 1$ holds, it may happen that groups choose to stay at the same section to next time point when the converging point is congested. Constraints ensure that one group will not stay at same section for more than three time points. In constraints, it is restricted that groups can stay at same section at next time point only if its connected sections are loaded with other groups. Constraints ensure that each group can be loaded on no more than one outbound truck. Constraints guarantee that parcels loaded onto each outbound truck is no more than its capacity limitation. Constraint ensure that a group can be put onto outbound truck o only if it arrives at the shipping dock door before the departure time of o . Constraint guarantee that a group can only be put onto outbound truck o docked at its determined shipping dock door. Constraints determine the amount of extra outbound trucks needed for delayed parcels. Constraint calculate the maximum finish time of transshipment among all groups.

4 Methodology

Boysen et al. (2013) showed truck scheduling with fixed outbound as a NP-hard problem, and therefore this problem can be solved to optimality only for small size problems. As a result, metaheuristic algorithms are developed to deal with this problem. Started from genetic algorithm, dynamic parameter settings and local search are also considered. This study applied adaptive genetic algorithms with local search (LSAGA) to solve the proposed problem. Section 4.1 described the framework of algorithm, and Sect. 4.2 elaborated the elements and processes of LSAGA.

4.1 Algorithm framework

An adaptive genetic algorithm with local search is proposed to improve the search ability of GA and achieve convergence efficiently. The flow chart of the LSAGA is shown in Fig. 9.

4.2 Steps of LSAGA

4.2.1 Chromosome representation

In genetic algorithm, each chromosome represents a solution to the problem. There were three sub-problems, sequence of inbound trucks, door assignment, and converging and diverging policy. Therefore, the chromosome is made up with three sub-chromosomes. The combination of first and second sub-chromosome, IS string and RS string, decides the inbound sequence of receiving dock doors. For these two sub-chromosomes, the encoding method is extended from Li and Gao (2016). The third

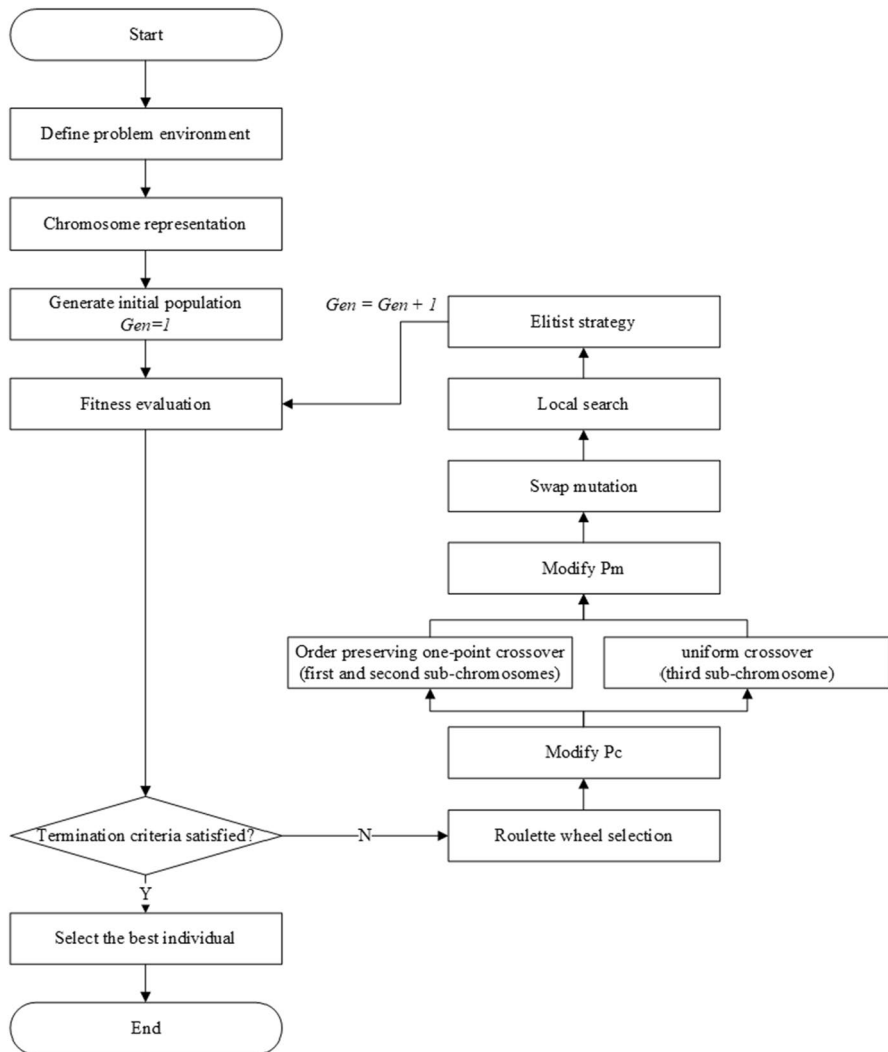


Fig. 9 Framework of LSAGA

sub-chromosome indicates diverging and converging policy for each diverging or converging point (see Fig. 10).

The encoding method of IS string is permutation encoding which presented the arrangement of given set of inbound trucks. Therefore, length of a single IS string must be the total number of inbound trucks with values not repeated. The RS string denotes the selected receiving dock door for corresponding inbound truck. The length of RS string is also the total number of inbound trucks. Value

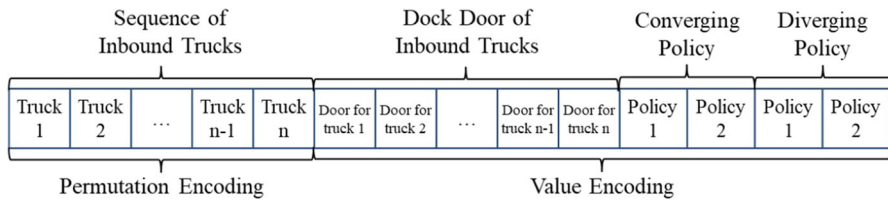


Fig. 10 Chromosome design

encoding was used to represent receiving dock doors as every gene in the string indicated the selected receiving dock door. Assuming that a conveyor system is with $|R|$ receiving dock doors and $|I|$ inbound trucks, then IS string is a permutation of $[1, |I|]$, and RS string is a string only contains values of $[1, |R|]$ with length i . An example with 2 receiving dock doors and 4 inbound trucks is provided in Section A1 of Appendix A. IS string, $\{3, 1, 2, 4\}$, corresponds to RS string, $\{1, 2, 1, 2\}$. Both I_3 and I_2 are assigned to R_1 ; meanwhile, I_1 and I_4 are assigned to R_2 . The sequence is decided by the precedence in IS string. I_3 is assigned to R_1 , and I_2 afterwards as I_3 is in front of I_2 in IS string.

The third sub-chromosome, PS string, assigns policy to each diverging and converging point. Assuming that a conveyor system is with cp converging points and dp diverging points, the first cp genes represents the policy to converging points respectively and the following dp genes are diverging policy to corresponding diverging points. Therefore, the length of PS string will be $cp + dp$. For PS string, value encoding was adopted to represent policy of corresponding converging and diverging point. For example, there are two diverging point, D1 and D2, and two converging point, C1 and C2, and the PS string is $\{3, 4, 2, 1\}$. The policy of C1 is the further from the destination goes first and lower loading rate goes first for C2. Groups choose to go to shortcut than go to main conveyor at D1, and groups choose to go to main conveyor at D2 (see Sect. A2 of Appendix A).

4.2.2 Generate initial population

The first step of genetic algorithm is to generate initial population. In this research, individuals are randomly generated for the initial population. Firstly, generate a series of consecutive numbers corresponding to the number of inbound trucks and shuffle them as first sub-chromosome. The second sub-chromosome is made up of random number of receiving dock doors, and the third sub-chromosome is also composed of random number of respective policies. Generally, a small population size may result in low diversity and cause premature convergence. However, a population size larger than necessary can waste of computational resource. This paper set population size as fifty.

4.2.3 Fitness value evaluation

As initial population is generated, next step is to evaluate fitness value of each individual. In this research, the objective is to minimize the summation of extra outbound truck cost and holing cost. Therefore, the total cost is observed and used to measure fitness value of individuals. Section A3 of Appendix A shows the process of fitness value evaluation. After decoding chromosome, we can generate inbound truck sequence for each receiving dock door and policy of each converging and diverging point. Detailed description is as follows.

1. Transfer chromosome into inbound truck sequences of receiving dock door. Set $X_{r,0,i}, X_{r,i,j}, X_{r,i,n+1}$ for $r \in R$.
2. Dock first inbound truck to receiving dock doors. Assign inbound truck i with $X_{r,0,i} = 1$ to receiving dock door r . Put first group onto the conveyor for each inbound truck i at receiving dock door r .
3. According to the fixed outbound truck schedule, dock and undock outbound trucks with arriving time and leaving time as time t . If there is any group in temporary storage, then put it onto outbound truck right after the truck is docked.
4. If any group k is on the section m connected with shipping dock door at time t , check if the destination of group k is exactly the door, i.e., $m = cm_{q_k}$. If yes, $ET_k = t$ and $Z_{k,cm_{q_k},ET_k} = 1$, then verify the remaining space of outbound truck. In the case of fully loaded outbound truck, groups are put in the temporary storage at the receiving dock door and wait for next outbound truck or extra truck; otherwise, groups are put onto the outbound trucks.
5. For converging points, decide which group can go. Assuming shortcut section a and conveyor section b converge into one section n of conveyor with $g_{a,n} = 1$ and $g_{b,n} = 1$, check if any group stay at the same section for two time periods. If $X_{k,a,t} = 1$ and $X_{k,a,t-1} = 1$, assign $X_{k,n,t+1} = 1$; otherwise, give the priority to both groups base on policy of the converging point.
6. For diverging points, check if any section m of diverging point is loaded as $g_{m,a} = 1$, $g_{m,b} = 1$, and $X_{k,m,t} = 1$. If yes, decide which route to go based on diverging policy. If the route with shortcut has higher priority, it is necessary to confirm if the destination of the group is reachable by using the shortcut.
7. Update the location of the remaining group k at section m which is not diverging or converging point. If the next section is loaded with group l , $g_{m,n} = 1$ and $X_{l,n,t+1} = 1$, the group stays at the same section, $X_{k,m,t+1} = 1$; otherwise, go to next section, $X_{k,n,t+1} = 1$.
8. For $r \in R$, if the section connected with receiving dock door r is empty, $\sum_{k \in K} X_{k,cm_r,t+1} = 0$, put a new group l onto the conveyor system from the inbound truck docked at receiving dock door r and assign $X_{l,cm_r,t+1} = 1$ and $ST_l = t$.
9. If inbound truck docked at receiving dock door r is empty, dock next inbound truck to receiving dock door r . Afterwards, go back to step 3 with $t + 1$.

If all parcels are transshipped, we can calculate total number of delayed parcels, $\sum_{k \in K_s} w_k \times (1 - \sum_{o \in O} H_{k,o})$, and subsequently determine N_s , the number of extra

outbound trucks needed for shipping dock door s . Finally, sum up the holding cost of each group. For groups put onto pre-determined outbound trucks, the holding cost is considered as $(\sum_{o \in O} d_o \times H_{k,o} - ET_k) \times hc$. On the other hand, as extra outbound trucks are dispatched right after the last group arrives at the shipping dock door, the holding cost is considered as $(ET_{\max} - ET_k) \times hc$ for groups put onto extra outbound trucks. The fitness value is obtained by adding up the cost of extra outbound trucks and holding cost.

4.2.4 Selection

After evaluating individuals, the selection of parents which mate to generate next generation was performed. Roulette wheel selection was applied to select parents to create offspring. In a roulette wheel selection, each individual in population has a proportion of the circular wheel, which is proportional to its fitness value. The better the fitness value, the greater chance for that individual to be chosen. Choosing individuals can be depicted as spinning a roulette. An individual is selected when the point hit in the roulette is contained in the proportion of the individual.

4.2.5 Crossover

After selecting parents, the selected chromosomes are recombined to create offspring. Two crossover operators are applied in this research. Order preserving one-point crossover operator is used for first and second sub-chromosomes, IS string and RS string, and uniform crossover operator is applied to the third sub-chromosome, PS string.

Order preserving one-point crossover firstly decides a cut point. The genes before the cut point are preserved to the offspring from parents, and then copy the remaining unused numbers from the second parent to the first child, wrapping around the list. Repeat the same action for the second child with the first parent. Since the inbound truck sequences of receiving dock doors are decided by the combination of IS string and RS string, the order preserving one-point crossover is conducted on both chromosomes with same cut point.

In a uniform crossover, each gene is chosen from either parent with equal probability. It treats both genes of two chromosomes independently to choose for offspring. A mask is randomly generated to decide the gene is taken from whether first parent or second parent. At each gene, if there is a 0 in the mask at the index, the gene of first offspring is copied from first parent, and the gene of second offspring is copied from second offspring; alternatively, if there is a 1 in the mask at the index, an interchange between gene from first parent chromosome and second parent chromosome is done to develop gene for first offspring chromosome and second offspring chromosome.

The flow chart of crossover is shown in Sect. A4 of Appendix A. The process of crossover is described as below:

1. Select 2 parent chromosomes by roulette wheel selection from current population.
2. Execute order preserving one-point crossover for IS string and RS string.

- i. Generate a number follows uniform distribution, $[1, |I|]$ as a cut point, which indicates the genes participating crossover. $|I|$ denotes the total amount of inbound trucks.
 - ii. The genes before cut point are transferred from first parent chromosome to first offspring chromosome. Repeat the same action for second parent and second offspring chromosome.
 - iii. Select genes in second parent chromosome which are not contained in genes of first parent chromosome before cut point, and then fill in first offspring chromosome in order. Repeat the same action for first parent and second offspring chromosome.
3. Execute uniform crossover for PS string.
 - i. Randomly generate a number between 0 and 1 for each gene. If the number is bigger than 0.5, then give 0 for the position to the mask; otherwise, give 1.
 - ii. Keep the gene from first parent to first offspring if there is a 0 in the mask and transfer the gene from second parent to first offspring if there is a 1 in the mask.
 4. Repeat step 1 to 3 until enough offspring chromosomes are generated.

Figure 11 shows an example of crossover with 8 inbound trucks, 4 receiving dock doors, 4 converging points with 5 policies, and 4 diverging points with 3 policies. In the example, the selected cut point was 4, so the genes of parent chromosomes before cut point 4 were copied to offspring chromosomes. Genes after cut point were filled by non-repetitive genes from another parent. After crossover of IS string and RS string, a mask was made for uniform crossover where 1 representing interchange and 0 representing maintain. Finally, the offspring chromosomes were generated.

4.2.6 Mutation

Mutation is an operator to maintain diversity of the population and avoid getting trapped into local optima. It alters one or more genes of original chromosome and therefore change the solution entirely. A better solution may sometimes come out by mutation operator. Swap mutation is employed to deal with the mutation in this research. Two swap operators are performed separately on IS string and RS + PS string because of the encoding method, permutation encoding for IS string and value encoding for RS and PS string. Since the amount of receiving dock doors, converge policies, and diverge policies are not the same, repair is essential for chromosomes after mutation. The process of mutation is shown as below:

1. For each chromosome in the population, generate a random number, $[0, 1]$. If the number is smaller than the mutation rate, the chromosome undergoes mutation.

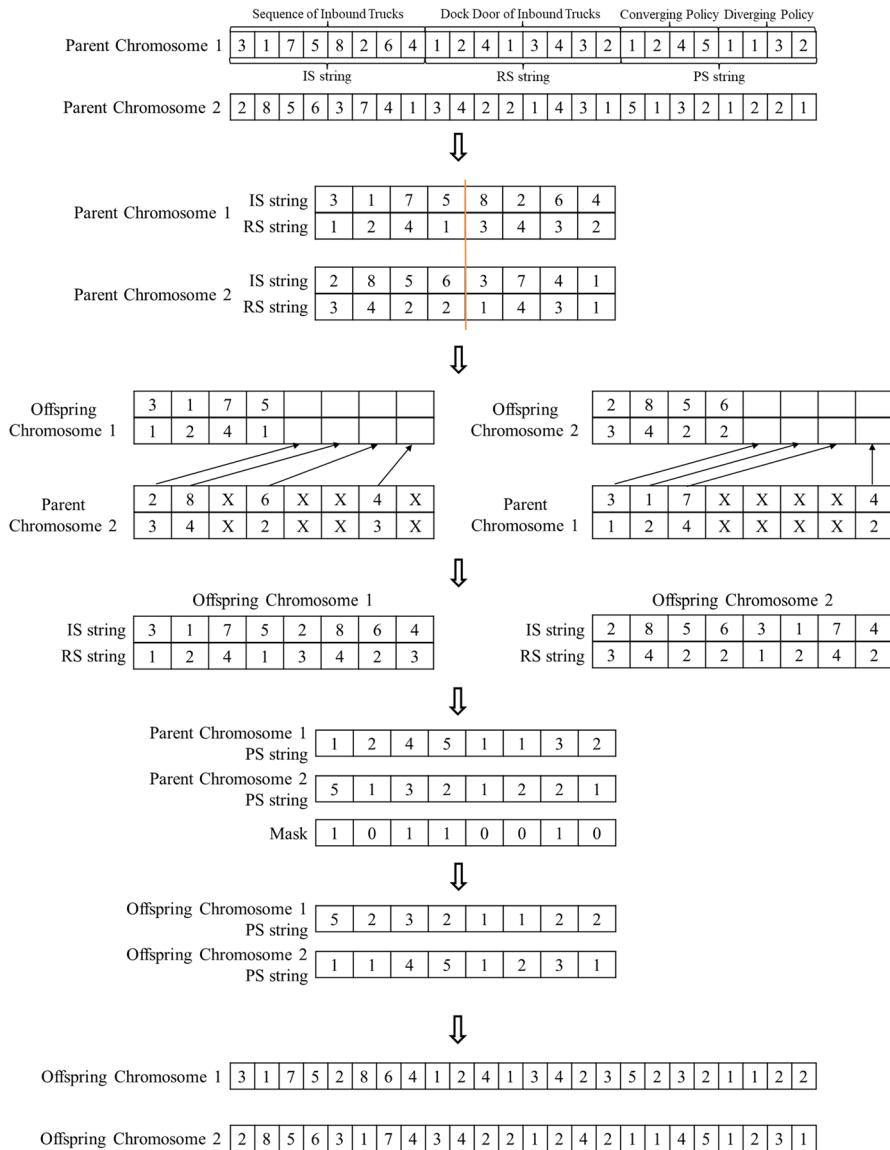


Fig. 11 An example of crossover

2. Randomly generate four integers a , b , c and d following discrete uniform distribution where $a, b \in [1, |I|]$ and $c, d \in [|I| + 1, L]$. $|I|$ denotes the total number of inbound trucks, and L denotes the length of the chromosome.
3. Interchange the genes at position a and b , and so do genes at position c and d .
4. Repair genes that is inappropriate by creating a random number between 1 and corresponding upper limit.

Figure 12 shows an example of mutation with 8 inbound trucks, 4 receiving dock doors, 4 converging points with 5 policies, and 4 diverging points with 3 policies. In this example, the random number, a , b , c and d , were generated as position 3, 7, 10, and 17. Therefore, genes at position 3 and 7 were interchanged, and so do genes at position 10 and 17. However, the gene at position 10, which represented the receiving dock door for inbound truck 1, was out of range. A random number that follows discrete uniform distribution $[1, |R|]$ was created to repair the gene. $|R|$ is the number of receiving dock doors.

4.2.7 Local search

Local search techniques use local information about current solution to reach a new solution within few modifications. It explores the neighborhood solutions of current selected chromosome to find a better solution. When a better solution is found by neighborhood exploration, the current solution is replaced. On the other hand, when there is no better solution found, the current solution remains unchanged. The local search techniques improve the search efficiency of GA and AGA and the quality of individuals.

The local search is turned on after the mutation operations. A certain proportion of individuals are randomly selected to undergo local search. An inbound truck is selected to redetermine the receiving dock door to find best fit dock door with lowest fitness value. Afterwards, we adapted the neighborhood operator from Bodnar et al. (2017) to exchange the sequence of the selected inbound truck with other inbound trucks assigned to the same receiving dock door and select the sequence with lowest

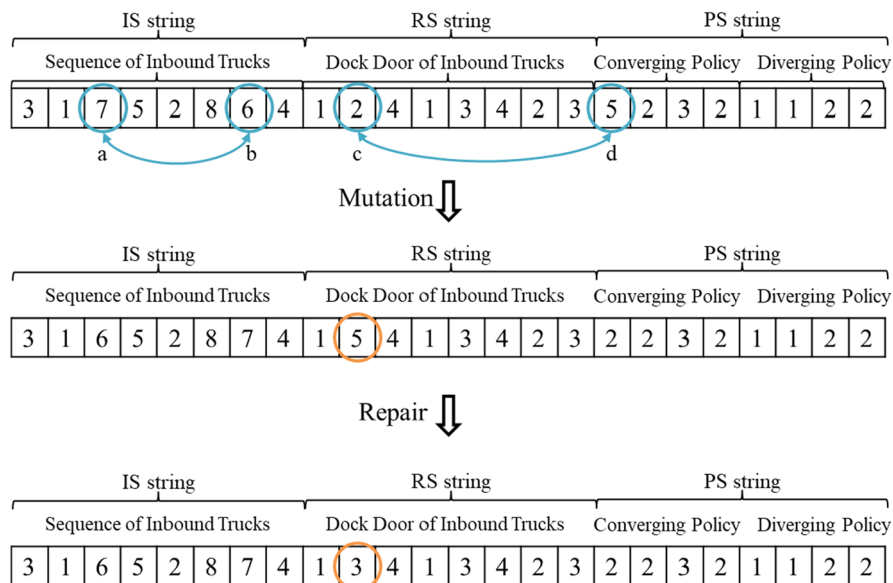


Fig. 12 An example of mutation

fitness value. A maximum search amount is set to avoid affecting computing efficiency for this part of search. One converging or diverging point is also selected to find the policy with best fitness. In this research, we set the local search rate as 10% and maximum local search amount as five of the searches of inbound sequence. Section A5 of Appendix A shows the flow chart of local search. The local search process is designed as follows:

1. For each chromosome in the population, generate a random number, $[0, 1]$. If the number is smaller than the local search rate, the chromosome undergoes local search.
2. Select one inbound truck i by generating a random number between 1 and $|I|$, the amount of inbound trucks.
3. Create neighborhood solutions by changing the receiving dock door of inbound truck i . If the fitness value of neighborhood solution is lower than the original one, replace the original solution with the neighborhood solution. Thus, we can obtain the best receiving dock door r for inbound truck i .
4. Exchange the sequence of the receiving dock door r of inbound truck i to create neighborhood solutions. If the fitness value is lower, change the arrangement of inbound trucks at receiving dock door r .
5. Generate a random number, $[2|I| + 1, L]$, to select one converging or diverging point p to do local search. $|I|$ denotes the amount of inbound trucks, and L denotes the length of chromosome.
6. Change the policy of p to explore neighborhood solutions, and obtain the best policy for p under current inbound truck arrangement.

Figure 13 shows an example of local search in this research. In this example, the inbound truck 6 was selected to explore neighborhood by random selection. Neighborhood solutions were generated by changing receiving dock door of inbound truck 6. The solution with receiving dock door 4 was the one with best fitness value, so it was chosen to conduct following local search. Next, inbound truck 6 was swapped with other trucks in the sequence of receiving dock door 4, inbound truck 8 and 7. The solution 4 with lower fitness value replaced solution 3 to next step. The second diverging point was selected to conduct local search. Since there were 3 possible diverge policies, the neighborhood of solution 4 were generated by changing to the original policy 2 to policy 1 and 3. The change of policy of diverging point 2 didn't bring out better solution, so solution 4 was then chosen to replace the original chromosome. The fitness value of this chromosome decreased from 1950 to 1650 after local search.

4.2.8 Replacement

In order to maintain the quality of the solution, 10% of the elite will be copied directly to the next generation based on elitist strategy, and another 90% of offspring are generated through crossover and mutation. After crossover and mutation are completed, 10% of non-elite offspring, will perform local search.

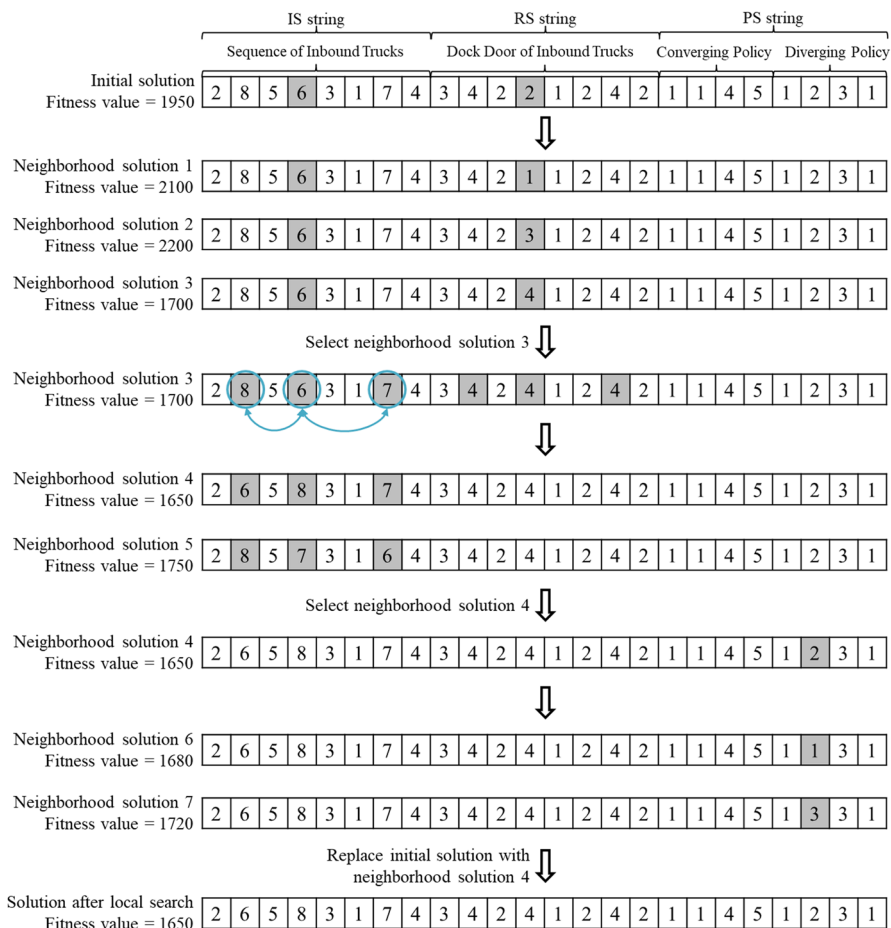


Fig. 13 An example of local search

4.2.9 Termination criteria

The termination criteria decide the conditions when GA will end. Normally, an ideal termination condition is when the algorithm reaches the approximate optimal solution. The operation of selection, crossover, mutation, and replacement will continue until the termination criteria are satisfied. In this research, the proposed algorithm will terminate when a fixed number of generations is reached.

4.2.10 Adaptive probabilities of crossover and mutation

Srinivas and Patnaik (1994) discussed that increasing p_c and p_m can promote exploration but also demote exploitation, and vice versa. It is suggested p_c and p_m be

increased when the population tend to stuck at a local optimum and be decreased when the population is scattered. The expressions of p_c and p_m are given as

$$p_c = \begin{cases} k_1 \times \left(\frac{f_{\max} - f'}{\bar{f} - f'} \right), & f' \geq \bar{f} \\ k_3, & f' < \bar{f} \end{cases} \quad (28)$$

and

$$p_m = \begin{cases} k_2 \times \left(\frac{f_{\max} - f}{\bar{f} - f} \right), & f \geq \bar{f} \\ k_4, & f < \bar{f} \end{cases} \quad (29)$$

where $k_1, k_2, k_3, k_4 \leq 1.0$

However, (28) and (29) apply to the maximization problem, but this research is considering a minimization problem. Therefore, a few minor adjustments were made to (28) and (29) where f' is the smaller of the fitness values of the parents selected to be crossed. The expressions of p_c and p_m are given as follows:

$$p_c = \begin{cases} k_1 \times \left(\frac{f' - f_{\min}}{\bar{f} - f_{\min}} \right), & f' \leq \bar{f} \\ k_3, & f' > \bar{f} \end{cases} \quad (30)$$

and

$$p_m = \begin{cases} k_2 \times \left(\frac{f - f_{\min}}{\bar{f} - f_{\min}} \right), & f \leq \bar{f} \\ k_4, & f > \bar{f} \end{cases} \quad (31)$$

where $k_1, k_2, k_3, k_4 \leq 1.0$

The adjusted rate should not exceed the range from 0.5 to 1.0 for p_c and the range from 0.001 to 0.05 for p_m . The solutions with subaverage fitness are expected to be completely disrupted to explore search space for the region containing global optimum. Therefore, the value of k_1 and k_3 is assigned as 1.0, and the value of k_2 and k_4 is assigned as 0.5.

5 Computational study

To evaluate the performance of the proposed LSAGA, a computational analysis is conducted in this section. In Sect. 5.1, a design of experiments (DOE) is carried out to verify significant factors. Section 5.2 covers the comparison among 4 algorithms: genetic algorithm (GA), adaptive genetic algorithm (AGA), genetic algorithm with local search (LSGA), and adaptive genetic algorithm with local search

(LSAGA). All experiments were implemented in C++ under Intel® Core™ i7-7700 @ 3.60 GHz computer with 48.0 GB RAM.

5.1 Design of experiments

A design of experiments was conducted to identify factors that have a significant effect on the fitness. An experiment was conducted by using full factorial experiments with four factors. All factors are described in detail in Sect. 5.1.1, and the results are presented in Sect. 5.1.2.

5.1.1 Factors of DOE

There were four factors with different levels considered for full factorial experiments, and all these factors were introduced in this section. Layout design influenced the total numbers of sections, shortcuts, receiving dock doors, and shipping dock door. Inbound truck was a three-level factor, with 12, 24, and 36 which represented the small, medium, and large scale of inbound trucks. Outbound truck was a two-level factor considering the amount of outbound trucks is enough or not. Algorithm was a four-level factor where four different algorithms were used. Table 4 shows a summary of the factors and levels of proposed experiments.

5.1.1.1 Layout Two types of layout of the conveyor system were used in the computational study. The first layout is called simple layout as shown in Fig. 14. Simple layout contains one main conveyor, four shortcuts, two receiving dock doors, and four shipping dock door. The length and width are twenty and eight. The main conveyor consists of fifty-six sections, and each shortcut consists of eight sections. The main conveyor can only transport parcels counterclockwise, and the shortcuts are one-way flow. The capacity of each section is set as ten parcels. A total of four converging points and four diverging points need to be decided on the policy. However, only the groups with specific destinations can enter shortcut to avoid infinite cycle.

The second layout is a U-shaped layout which is called complex layout (see Fig. 15). Complex layout contains one main conveyor, six shortcuts, four receiving dock doors, and six shipping dock door. The length and width are twenty-four and sixteen. The main conveyor consists of one hundred sections, and each shortcut consists of six sections. The main conveyor can only transport parcels counterclockwise, and the shortcuts are one-way flow. The capacity of each section is set as ten parcels. A total of six converging points and six diverging points need

Table 4 Factors and levels for DOE

Factors	Levels			
	Simple	Complex		
Inbound truck	12	24	36	
Outbound truck	Sufficient	Insufficient		
Algorithm	GA	AGA	LSGA	LSAGA

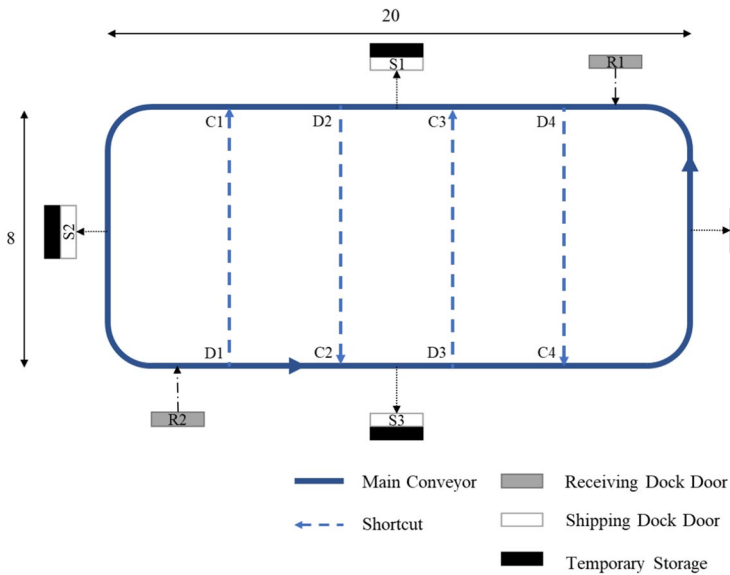


Fig. 14 Illustration of the simple layout

to be decided on the policy. Only the groups with specific destinations can enter shortcut to avoid infinite cycle. Table 5 shows the comparison of simple layout and complex layout.

5.1.1.2 Inbound truck Three different numbers of inbound trucks were designed for simple layout and complex layout, as shown in Table 6. Scenarios with 12, 24, and 36 inbound trucks were considered for both simple and complex layout. The setup time was set as one time period for each inbound truck. The number of parcels for different destinations in each inbound truck was generated from discrete uniform distribution on the interval $[0, 50]$. Since the numbers of destinations of simple layout and complex layout were 4 and 6, the average numbers of parcels in each inbound trucks of simple and complex layout were about 100 and 150, respectively.

5.1.1.3 Outbound truck Outbound trucks have a predetermined arrival time and departure time in this research. Two levels of outbound trucks were utilized in the DOE which were sufficient and insufficient. The capacity of outbound truck was fixed at 100. Table 7 shows the detail of parameter designed for outbound trucks. As for simple layout, the number of outbound trucks was set 12, 24, and 36 according to the total amount of parcels in inbound trucks in the scenarios with inadequate outbound trucks, and the number of outbound trucks was set 16, 28, and 40 in the scenarios with adequate outbound trucks. Similarly, for complex layout, the number of outbound trucks was set 18, 36, and 54 in the scenarios with inadequate outbound trucks, and the number of outbound trucks was set 24, 42, and 60 in the scenarios with adequate outbound trucks.

Table 6 Detail of inbound trucks

Layout	Number of destinations	Number of inbound trucks	Total amount of parcels	Average amount of parcels in each inbound truck
Simple	4	12	1228	102.33
		24	2439	101.63
		36	3664	101.78
Complex	6	12	1808	150.67
		24	3603	150.13
		36	5617	156.03

Table 7 Detail of outbound trucks

Layout	Number of inbound trucks	Total amount of parcels	Number of outbound trucks	
			Insufficient	Sufficient
Simple	12	1228	12	16
	24	2439	24	28
	36	3664	36	40
Complex	12	1808	18	24
	24	3603	36	42
	36	5556	54	60

Table 8 Comparison of processes and mechanisms between algorithms

Process	Mechanism			
	GA	AGA	LSGA	LSAGA
Random generation of initial population	✓	✓	✓	✓
Evaluation	✓	✓	✓	✓
Selection	✓	✓	✓	✓
Crossover operator	✓	✓	✓	✓
Mutation operator	✓	✓	✓	✓
Elitist Strategy	✓	✓	✓	✓
Adaptive crossover rate	✗	✓	✗	✓
Adaptive mutation rate	✗	✓	✗	✓
Local search	✗	✗	✓	✓

ing chromosomes, initial population generation, evaluation, selection, crossover, mutation, and replacement with elitist strategy. The operators of crossover and mutation are identical. The main differences among these algorithms are the local search which explore neighborhood and the adaptive adjustment of probabilities of crossover and mutation.

Table 9 Parameter settings for each algorithm

Algorithm	GA	AGA	LSGA	LSAGA
Population size	50	50	50	50
Generation size	400	400	400	400
Crossover rate	0.85	0.85	–	–
Mutation rate	0.05	0.05	–	–
Elitism rate	0.10	0.10	0.10	0.10
Local search rate	–	–	0.10	0.10
k_1	–	1.00	–	1.00
k_2	–	0.50	–	0.50
k_3	–	1.00	–	1.00
k_4	–	0.50	–	0.50

Table 10 ANOVA table for the DOE

Source	DF	F-Value	P-Value
Layout	1	5688.23	0.000
Inbound	2	10,197.96	0.000
Outbound	1	105.67	0.000
Algorithm	3	56.10	0.000
Layout*Inbound	2	1090.11	0.000
Layout*Outbound	1	102.13	0.000
Layout*Algorithm	3	1.49	0.215
Inbound*Outbound	2	32.12	0.000
Inbound*Algorithm	6	20.32	0.000
Outbound*Algorithm	3	1.66	0.175
Error	935		
Total	959		

$S = 1381.09$ R-Sq = 96.86% R-Sq (adj) = 96.78%

The parameter settings for each algorithm are shown in Table 9. The population size of the four GAs is fifty, and the generation size is 400. The crossover rate and mutation rate are 0.85 and 0.05 for GA and LSGA and remain constant from beginning to end. However, for AGA and LSAGA, the crossover rate and mutation rate are calculated from Eq. (30) and, so they vary according to selected chromosomes. Ten percent of elitist are preserved to next generation for four algorithms. Local search rate is set to 0.1 for LSGA and LSAGA.

5.1.2 Results of DOE

A general full factorial design was conducted by the software Minitab. Every experiment was replicated twenty times which led to a total of 960 experiments for the analysis. Table 10 shows that the experiments suggested that all factors and most of the interactions between two factors be significant under 95%

confidence interval, including layout, inbound truck, outbound truck, and algorithm. Section A6 of Appendix A shows the residual plots for total cost. It indicates that the data followed normal distribution from normal probability plot and histogram. The assumption of equal variance and independency is also verified as there was no obvious trend in the plots on the right-hand side.

Figure 16 shows the main effects of all factors. It is indicated that inbound truck had the most significant impact on total cost. The larger the number of inbound trucks, the higher the total cost. The main reason may be that larger number of inbound trucks contains more parcels to be transshipped. The cost was also influenced by layout to some degree. The total cost was lower in simple layout. In simple layout, the number of shipping dock doors was lower, and there were fewer types of parcels in inbound trucks. The transshipment time can be reduced as the route may be shorter from receiving to shipping dock door in simple layout. The amount of outbound trucks also influenced the total cost because insufficient number of outbound trucks may increase the cost of extra outbound trucks. Algorithms applied to solve the proposed model also affected the total cost generated. LSAGA can obtain relatively better solutions than GA, AGA, and LSGA. In the next section, a comparison of the performance of different algorithms in different scenarios is shown to find the most suitable algorithm for this problem.

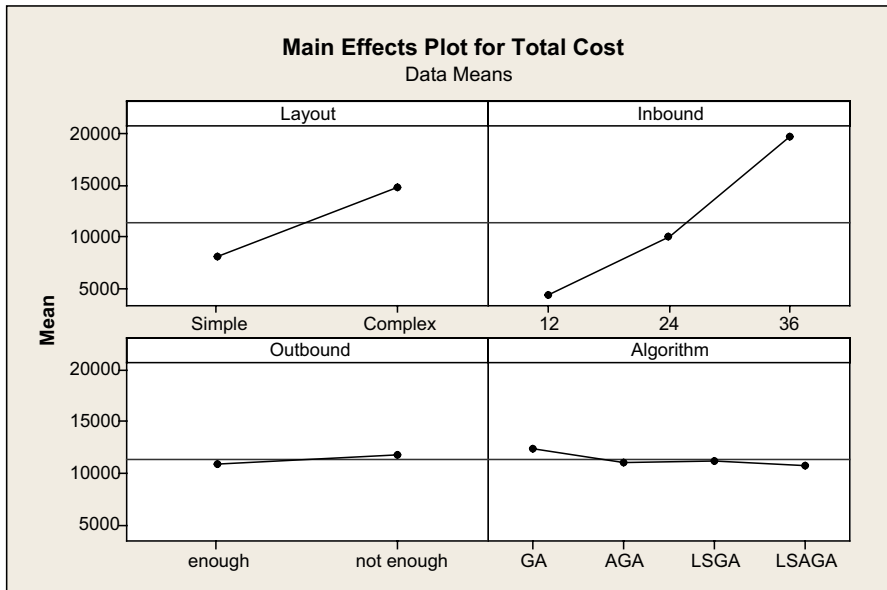


Fig. 16 Main effects plot for total cost

5.2 Performance comparison of different types of GAs

Algorithm is concluded as a significant factor to the objective value in previous section. In this section, we aim to compare four GAs and distinguish the best algorithm among these four for this problem. The key performance indices used in this section are confidence interval, box plot, convergence condition, run time, and the quality ratio. Twelve scenarios were designed, and twenty replications were run for each of them. Table 11 shows the detail of each scenario.

5.2.1 Objective value comparison

Table 12 shows the performance results, including the average of objective values, standard deviation of objective values, 95% confidence interval, maximum, and minimum over twenty replications for each algorithm under each scenario. The average objective value generated from LSAGA was the smallest except for scenario 7. The standard deviation of LSAGA was the lowest among algorithms under most scenarios, and it made the confidence interval smaller than others. The confidence interval became larger when the number of inbound trucks gets larger. Also, according to Table 15, LSAGA can obtain smaller objective value than other algorithms under most scenarios. As a result, we can conclude that LSAGA is stabler than other algorithms.

5.2.2 CPU time

The comparison of average CPU time over twenty runs executing four algorithms, GA, AGA, LSGA, and LSAGA is presented in Table 13. The average CPU time using GA operation is the shortest in all scenarios. The mechanism of adaptive adjustment of crossover rate and mutation rate was added to AGA to improve

Table 11 Details of each scenario

Scenario	Layout	Number of inbound trucks	Outbound trucks (number of outbound trucks)
Scenario 1	Simple	12	Enough (12)
Scenario 2		24	Enough (24)
Scenario 3		36	Enough (36)
Scenario 4		12	Not enough (16)
Scenario 5		24	Not enough (28)
Scenario 6		36	Not enough (40)
Scenario 7	Complex	12	Enough (18)
Scenario 8		24	Enough (36)
Scenario 9		36	Enough (54)
Scenario 10		12	Not enough (24)
Scenario 11		24	Not enough (42)
Scenario 12		36	Not enough (60)

Table 12 Performance results of each algorithm

Algorithm	Mean	SD	CI (95%)	Max	Min
<i>Scenario 1</i>					
GA	3262.80	113.60	(3035.59, 3490.01)	3517	3037
AGA	3191.90	99.57	(2992.77, 3391.03)	3435	3053
LSGA	3150.65	140.83	(2869.00, 3432.30)	3424	2947
LSAGA	3111.55	68.86	(2973.82, 3249.28)	3199	2981
<i>Scenario 2</i>					
GA	7623.60	259.04	(7105.52, 8141.68)	8211	7187
AGA	7548.05	177.85	(7192.36, 7903.74)	7935	7281
LSGA	7414.75	336.95	(6740.85, 8088.65)	8009	6895
LSAGA	7295.25	215.17	(6864.91, 7725.59)	7631	6906
<i>Scenario 3</i>					
GA	16,103.70	5165.18	(5773.34, 26,434.06)	26,365	12,403
AGA	12,303.95	349.17	(11,605.60, 13,002.30)	12,898	11,654
LSGA	13,259.40	3044.06	(7171.28, 19,347.52)	22,211	11,612
LSAGA	12,197.35	344.48	(11,508.39, 12,886.31)	12,803	11,526
<i>Scenario 4</i>					
GA	3171.55	154.43	(2862.69, 3480.41)	3587	2980
AGA	3058.55	130.54	(2797.48, 3319.62)	3306	2840
LSGA	3047.90	113.59	(2820.71, 3275.09)	3232	2824
LSAGA	2990.85	88.36	(2814.13, 3167.57)	3224	2858
<i>Scenario 5</i>					
GA	7645.95	205.39	(8056.74, 7235.16)	8112	7311
AGA	7413.50	176.99	(7767.47, 7059.53)	7805	7113
LSGA	7267.55	270.91	(7809.38, 6725.72)	7697	6787
LSAGA	7143.20	201.35	(7545.89, 6740.51)	7428	6769
<i>Scenario 6</i>					
GA	15,654.55	4669.63	(6315.29, 24,993.81)	25,443	12,357
AGA	12,618.65	468.36	(11,681.93, 13,555.37)	13,618	11,966
LSGA	14,278.80	3465.44	(7347.93, 21,209.67)	22,807	11,993
LSAGA	12,357.80	466.24	(11,425.32, 13,290.28)	13,588	11,617
<i>Scenario 7</i>					
GA	5470.05	192.25	(5085.55, 5854.55)	5871	5004
AGA	5318.50	221.48	(4875.54, 5761.46)	5628	4846
LSGA	5129.30	155.33	(4818.63, 5439.97)	5400	4917
LSAGA	5145.05	171.37	(4802.31, 5487.79)	5375	4809
<i>Scenario 8</i>					
GA	13,864.00	625.03	(12,613.94, 15,114.06)	15,116	12,919
AGA	11,755.50	343.00	(11,069.49, 12,441.51)	12,699	11,248
LSGA	11,538.50	397.98	(10,742.55, 12,334.45)	12,351	10,817
LSAGA	11,335.75	283.85	(10,768.04, 11,903.46)	11,924	10,511
<i>Scenario 9</i>					
GA	26,096.05	2102.08	(30,300.21, 21,891.89)	32,706	23,335

Table 12 (continued)

Algorithm	Mean	SD	CI (95%)	Max	Min
AGA	23,563.20	1081.29	(25,725.78, 21,400.62)	25,493	22,007
LSGA	23,873.15	1089.86	(26,052.88, 21,693.42)	25,257	20,798
LSAGA	23,246.45	889.07	(25,024.59, 21,468.31)	24,472	21,395
Scenario 10					
GA	6207.55	198.62	(5810.30, 6604.80)	6706	5868
AGA	5970.10	155.27	(5659.56, 6280.64)	6252	5616
LSGA	5887.80	206.97	(5473.86, 6301.74)	6416	5552
LSAGA	5815.75	132.24	(5551.26, 6080.24)	6173	5553
Scenario 11					
GA	13,808.90	620.48	(12,567.94, 15,049.86)	15,173	12,685
AGA	13,232.60	438.06	(12,356.48, 14,108.72)	14,055	12,415
LSGA	13,372.90	545.70	(12,281.49, 14,464.31)	14,206	12,484
LSAGA	12,774.85	401.60	(11,971.65, 13,578.05)	13,388	12,003
Scenario 12					
GA	29,533.10	1647.60	(26,237.91, 32,828.29)	33,187	26,172
AGA	27,538.25	1077.56	(25,383.13, 29,693.37)	30,110	26,152
LSGA	27,299.15	1352.43	(24,594.30, 30,004.00)	30,698	25,612
LSAGA	26,702.55	750.99	(25,200.57, 28,204.53)	27,769	25,335

Table 13 Average CPU time (in seconds) for each algorithm

Scenario	Average CPU time for algorithm (s)			
	GA	AGA	LSGA	LSAGA
Scenario1	44.222	68.902	115.587	139.200
Scenario2	220.470	301.195	553.774	639.745
Scenario3	322.486	440.483	622.988	718.837
Scenario4	165.106	248.532	321.992	426.459
Scenario5	206.453	320.415	543.596	625.600
Scenario6	221.452	415.132	606.537	824.881
Scenario7	364.961	662.891	943.677	1838.861
Scenario8	1191.577	1296.799	1946.010	3058.209
Scenario9	1505.484	2114.327	2736.710	4192.157
Scenario10	635.909	894.679	1520.121	1887.405
Scenario11	1178.966	1690.220	2827.725	4322.437
Scenario12	1970.890	2836.672	4351.017	5139.798
Average	668.998	940.854	1424.144	1984.466

performance of solutions and therefore increased the CPU time. Likewise, the CPU time was increased for LSGA with the mechanism of local search. For LSAGA, the CPU times were the highest under all scenarios since both

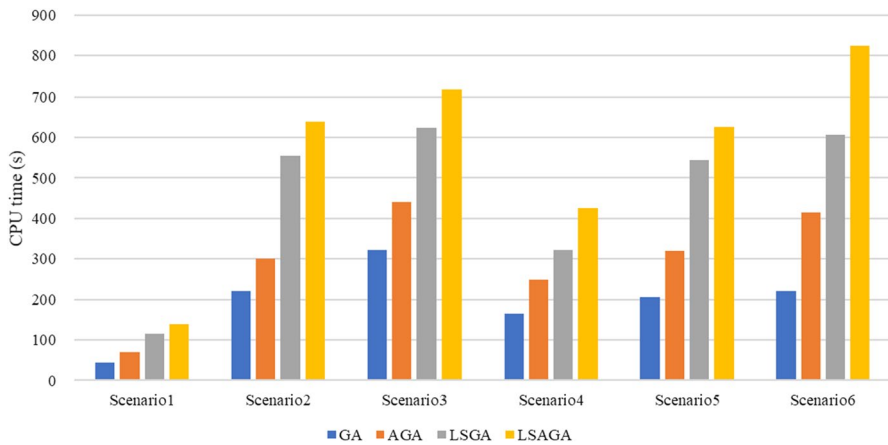


Fig. 17 Average CPU time of for algorithms under scenarios with simple layout

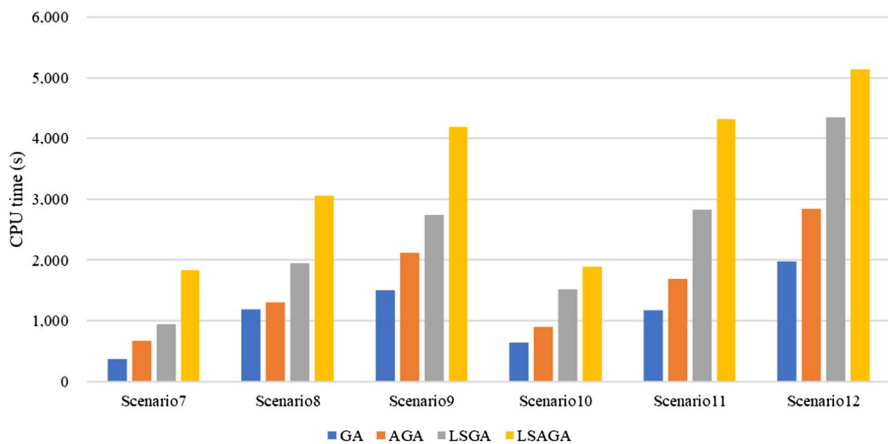


Fig. 18 Average CPU time of for algorithms under scenarios with complex layout

mechanisms were added. Figures 17 and 18 show the comparison of CPU time for algorithms under each scenario.

5.2.3 Convergent condition

When the fitness value no longer improves significantly, we call it convergence in general. In this study, local search and adaptive adjustment of p_c and p_m were added and expected to help GA break away from local optima and find a better fitness value. As a result, the convergent conditions of AGA, LSGA, and LSAGA may be different from traditional GA. Therefore, the convergent condition among four GAs is also an important performance index for computational study. The average fitness value of each generation was calculated to observe the

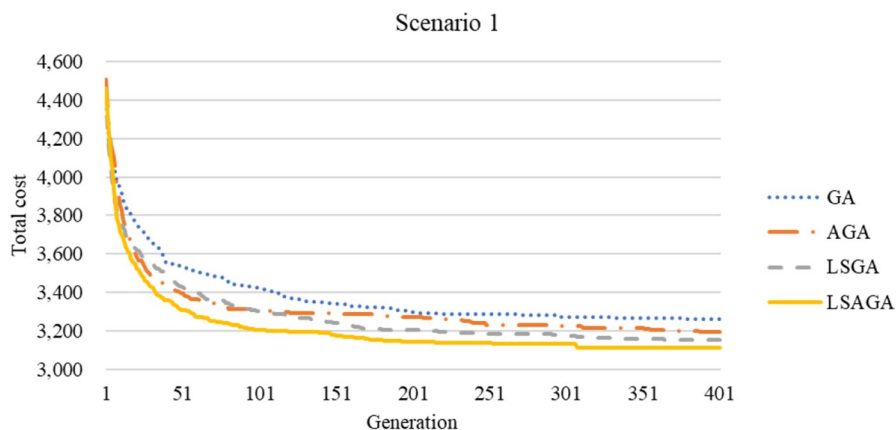


Fig. 19 Convergence plot under scenario 1

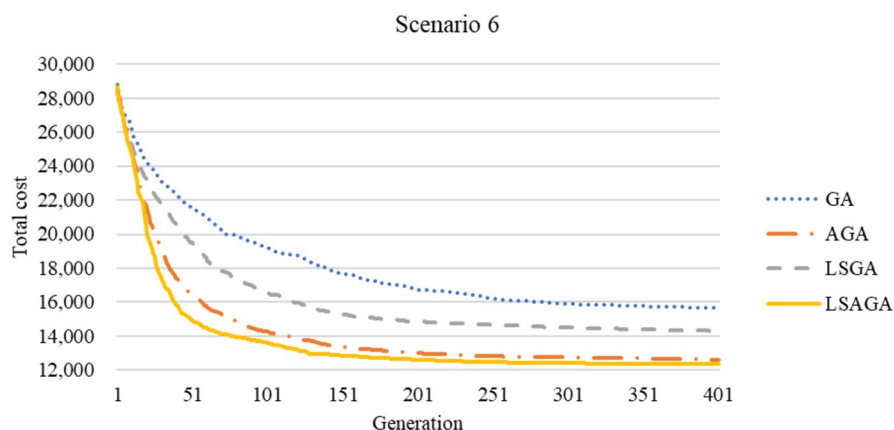


Fig. 20 Convergence plot under scenario 6

convergence process for each algorithm. As shown in Figs. 19, 20 and 21, GA converged to the worst solution among four algorithms in all scenarios. On the other hand, LSAGA converged to the best solution in most scenarios except scenario 7 where the solution was also close the best solution. The solutions obtained by LSAGA were better than GA, AGA, and LSGA after few generations in most scenarios. The performance of AGA and LSGA was unstable. AGA performed better than LSGA under some scenarios, and LSGA performed better than AGA in others. Convergence plot of all scenarios are shown in Appendix B.

LSAGA utilized fewer iterations to obtain better solutions among algorithms, and it converged to better solution than other solution under most scenarios. Therefore, we can conclude that LSAGA was the algorithm with better performance than other algorithms, and the mechanisms of local search and adaptive

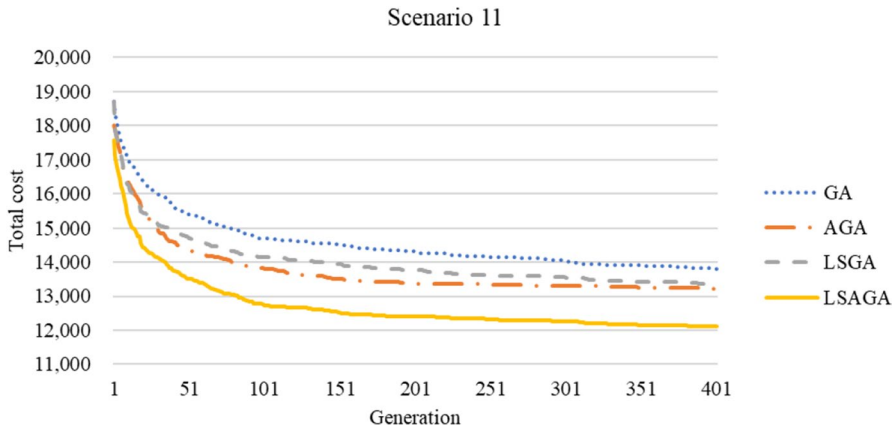


Fig. 21 Convergence plot under scenario 11

adjustment of p_c and p_m did help GA to escape from local optima and find better solutions.

5.2.4 Quality ratio

In this research, the quality ratio CTR proposed by Chen et al. (2019) is adapted to evaluate quality of algorithms. CTR is a ratio of the fitness value acquired by a specific algorithm over the minimum fitness value acquired by four algorithms. The quality ratio CTR is calculated as

$$CTR(a) = \frac{CT(a, s)}{CT^m(s)} \quad \forall a \in A \quad (32)$$

where A denotes the set of algorithms, $CT(a, s)$ denotes the average fitness value obtained by algorithm a on scenario s , and $CT^m(s)$ denotes the minimum of average fitness value obtained among all algorithms.

Table 14 shows the quality ratio CTR of four algorithms in different scenarios. The proposed LSAGA was outstanding in most scenarios, except for scenario 7. However, the CTR of LSAGA was very close to 1 which represented that the solution generated from LSAGA was closed to the minimum solutions. The average fitness value of all scenarios were used as $CT(a, s)$ to calculate the overall result of $CTR(a)$. The solutions obtained by LSAGA was 14.1%, 2.6%, and 4.2% better than the solutions obtained by GA, AGA, and LSGA, respectively. Therefore, the robustness of LSAGA is verified as LSAGA surpassed other GAs.

5.3 Sensitivity analysis

In this section, we conducted experiments to analyze the impact of the ratio of the cost of extra truck and holding cost to this problem. Also, outbound truck capacity

Table 14 Experiment results for CTR

Scenario	Algorithm			
	GA	AGA	LSGA	LSAGA
Scenario 1	1.049	1.026	1.013	1.000
Scenario 2	1.045	1.035	1.016	1.000
Scenario 3	1.320	1.009	1.087	1.000
Scenario 4	1.060	1.023	1.019	1.000
Scenario 5	1.070	1.038	1.017	1.000
Scenario 6	1.267	1.021	1.155	1.000
Scenario 7	1.066	1.037	1.000	1.003
Scenario 8	1.223	1.037	1.018	1.000
Scenario 9	1.123	1.014	1.027	1.000
Scenario 10	1.067	1.027	1.012	1.000
Scenario 11	1.081	1.036	1.047	1.000
Scenario 12	1.106	1.031	1.022	1.000
Overall	1.141	1.026	1.042	1.000

and scheduling method of outbound trucks are considered to analyze the effect on total cost. All the experiments are based on the scenario 6. LSAGA was used to obtain the result of experiments, i.e., fitness value, number of extra trucks, and total holding time. Twenty replications were run for each of experiments.

5.3.1 Cost ratio

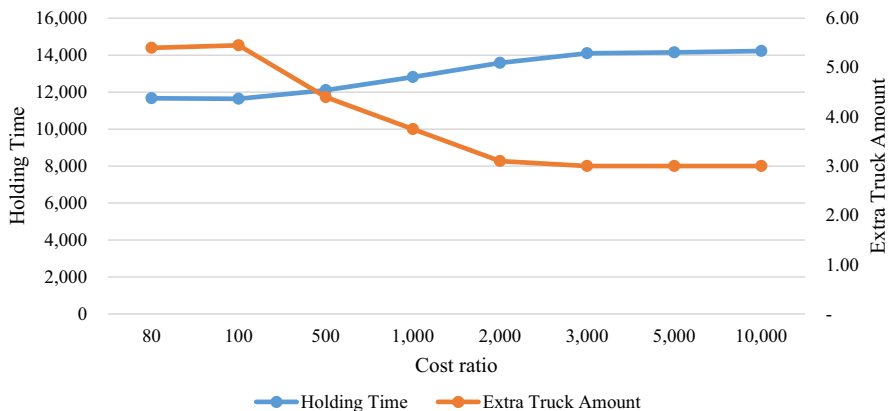
We considered different ratio of the cost of extra truck and holding cost. As total cost is defined as $\sum_{s \in S} N_s \times tc + \sum_{k \in K} ((\sum_{o \in O} d_o \times H_{k,o} + (1 - \sum_{o \in O} H_{k,o}) \times ET_{\max}) - ET_k) \times hc$, we can define the cost ratio as (33).

$$\text{cost ratio} = \frac{\text{extra truck cost}}{\text{holding cost}} = \frac{tc}{hc} \quad (33)$$

Since the extra truck cost is usually higher than holding cost, the cost ratios experimented are set higher than one. High cost ratio represented the extra truck cost is much higher than the holding cost, and low cost ratio represented the extra truck cost is close to holding cost. We went through different cost ratio and observed the change of extra truck amount and holding time. We can conclude that the cost ratio did influence the strategy of inbound truck scheduling. Table 15 and Fig. 22 showed that the scheduling result tends to choose to let parcels send latter and avoid parcels waiting at temporary storage area if the cost ratio is low, so that parcels will not stay at temporary storage area for a long time and reduce holding cost. However, as the cost ratio rises, the scheduling system takes another strategy where the parcels arrived at the shipping dock door in advance to catch the predefined outbound trucks and avoid the high cost of extra trucks.

Table 15 Sensitivity analysis of cost ratio

Cost ratio	80	100	500	1000	2000	3000	5000	10,000
Holding time	11,666.80	11,643.00	12,105.50	12,818.35	13,580.90	14,105.20	14,148.65	14,227.55
Extra truck amount	5.40	5.45	4.40	3.75	3.10	3.00	3.00	3.00

**Fig. 22** Total holding time and extra truck amount with different cost ratios**Table 16** Sensitivity analysis of truck capacity

Outbound truck capacity	60	80	100	120	150	200
Extra truck amount	39.00	24.10	5.45	3.05	1.15	0.00
Total cost	38,452.00	31,589.35	12,188.00	9,970.25	9,678.30	9,631.60

5.3.2 Outbound truck capacity

In this experiment, we considered scenario 6 where the cost ratio was 1000 and compared the total cost with different outbound truck capacity. As we can see from Table 16 and Fig. 23, the amount of extra trucks decreases with the increase of outbound truck capacity which in turn decreases the total cost. It is because that the parcels may be put in the temporary storage and wait for next outbound truck or extra trucks at the receiving dock door if the outbound truck at the dock door is full. If the capacity of outbound trucks increases, the parcels are with higher possibility to catch the predefined outbound trucks; moreover, lower extra trucks will be used for delayed parcels. In scenario 6, 36 outbound trucks were predefined, and 3664 parcels in total needed to be transported. If the outbound truck capacity is lower than 100, the predefined outbound trucks are obviously not sufficient for transportation, and therefore the cost of extra truck will be very high. When the capacity came to 200, no extra truck was required.

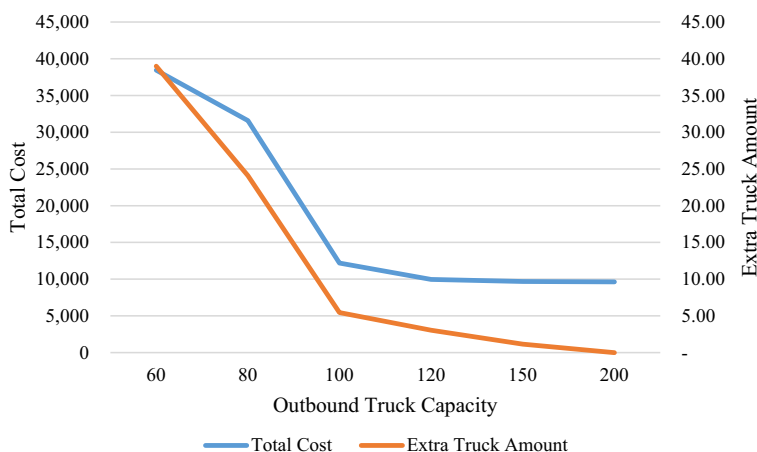


Fig. 23 Improvement obtained with different truck capacity

5.3.3 Scheduling method of outbound trucks

Different arrangement of outbound trucks is compared in this section. 36 outbound trucks were predefined in scenario 6. Three experiments are conducted where the total amount of outbound trucks remained the same, 36, but with different scheduling method. For the first method, outbound trucks were high-frequency during the first half period of time and low-frequency during the second half period of time. For the second method, the outbound trucks are equally distributed, i.e., the interarrival time is the same from beginning to end. For the third method, the scheduling method is opposite to the first experiment, so outbound trucks were low-frequency during the first half period of time and high-frequency during the second half period of time. Figure 24 shows an example of three scheduling methods of outbound trucks, and four outbound trucks are assigned to every time slice.

The result shows that method 2 is best of the three methods since the total cost of method 2 is the lowest. From Table 17 and Fig. 25, we can know that the holding cost of method 1 and 3 was relatively high, so that the total cost was higher than method 2. It was because that it took more time for parcels to wait for next outbound trucks during the time period with low-frequency. For method 1, outbound trucks were high-frequency during the first half period. However, it takes time for parcels transporting from shipping dock door to receiving dock door, so parcels may not arrive in the early stage. As a result, average loading rate of outbound trucks was lower in method 1, and more extra trucks were used. We can therefore conclude that the scheduling method with equally distributed outbound trucks is the most suitable method for this problem.

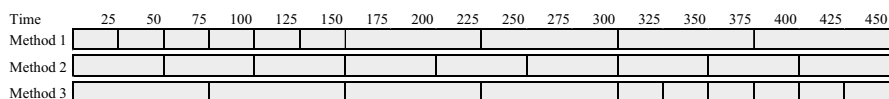
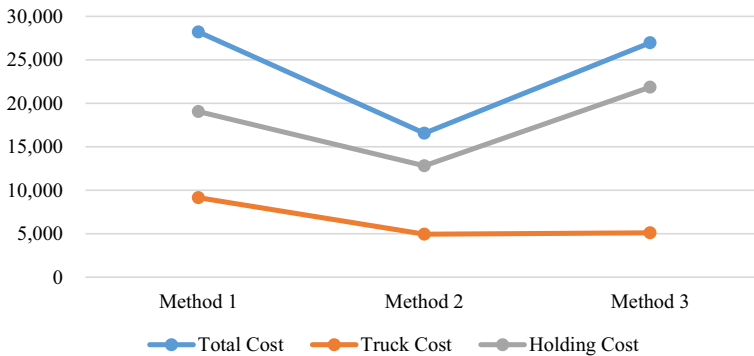


Fig. 24 Example of three scheduling methods of outbound trucks

Table 17 Sensitivity analysis of scheduling method of outbound trucks

	Method 1	Method 2	Method 3
Extra truck cost	9150.00	4950.00	5100.00
Holding cost	19,056.40	12,818.35	21,869.10
Total cost	28,206.40	16,568.35	26,969.10

**Fig. 25** Comparison of three scheduling methods of outbound trucks

6 Conclusion

With the development of the transportation industry, a logistics center is often used to carry out all the activities relating to transport, logistics, and goods distribution. Moreover, the time of transshipment from receiving dock door to shipping dock door of the logistics center is a major factor affecting the delivery time. However, truck scheduling with a fixed outbound schedule is often used in logistics centers such as post offices. In this study, we investigate a truck scheduling problem in a closed-loop conveyor system with shortcuts where the outbound departures are predetermined. The existence of shortcuts provides faster alternative routes to parcels that need to be transshipped. There are more than one routes from a receiving dock door to shipping dock door. A mixed integer nonlinear programming model is developed to minimize the cost of extra truck used to transport delayed parcels as well as holding cost of groups at shipping dock doors. Four GAs, GA, AGA, LSGA, and LSAGA, are used to solve the model due to the NP-hardness of proposed model. LSAGA is the proposed algorithm with local search and adaptive adjustment of crossover rate and mutation rate based on selected chromosomes to improve the solution quality. A DOE with four factors was conducted and showed that layout, inbound truck, outbound truck, and algorithm are all significant factors to the total cost. The comparison among four algorithms is conducted with key performance indices, confidence interval, box plot, CPU time, convergence condition, and quality ratio. Sensitivity analysis is performed to evaluate the influence of cost ratio, outbound truck capacity, and scheduling method of outbound trucks. Experiment results shows that adaptive adjustment and local search can help to improve GA, and LSAGA can obtain better solutions with stability.

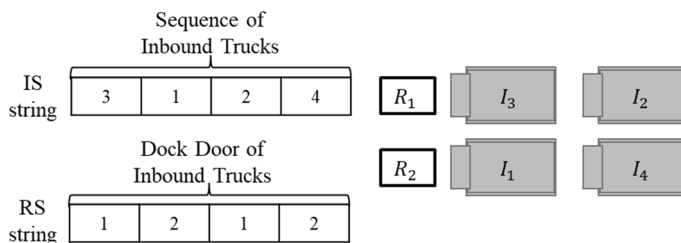
The contribution of this research is to define and solve the truck scheduling problem with fixed outbound departure in a closed-loop conveyor with shortcuts with efficient methodologies. Several factors are proved to be significant to this problem, and mechanisms are added to algorithms to take a balance between exploration and exploitation in the search process of genetic algorithm. Some managerial insights are also proposed to this problem.

This topic can be extended in following directions:

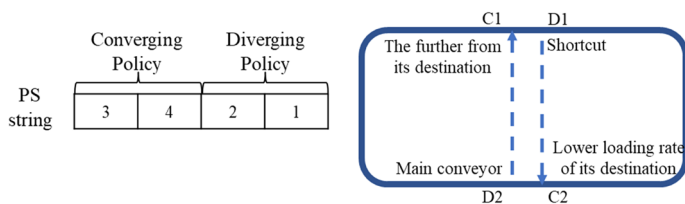
- This research considers inbound trucks arriving at the sortation system at the beginning of scheduling horizon. However, it will be closer to reality to consider inbound trucks with uncertain arrival times.
- The sequence of unloading parcels onto conveyor is also a significant factor to transshipment. Parcels are randomly selected to be unloaded in this research. Considering the order of unloading parcels into scheduling is worthy of being discussed.
- Improve the proposed LSAGA in this research or combine with other algorithms to enhance the quality and robustness of results.

Appendix A

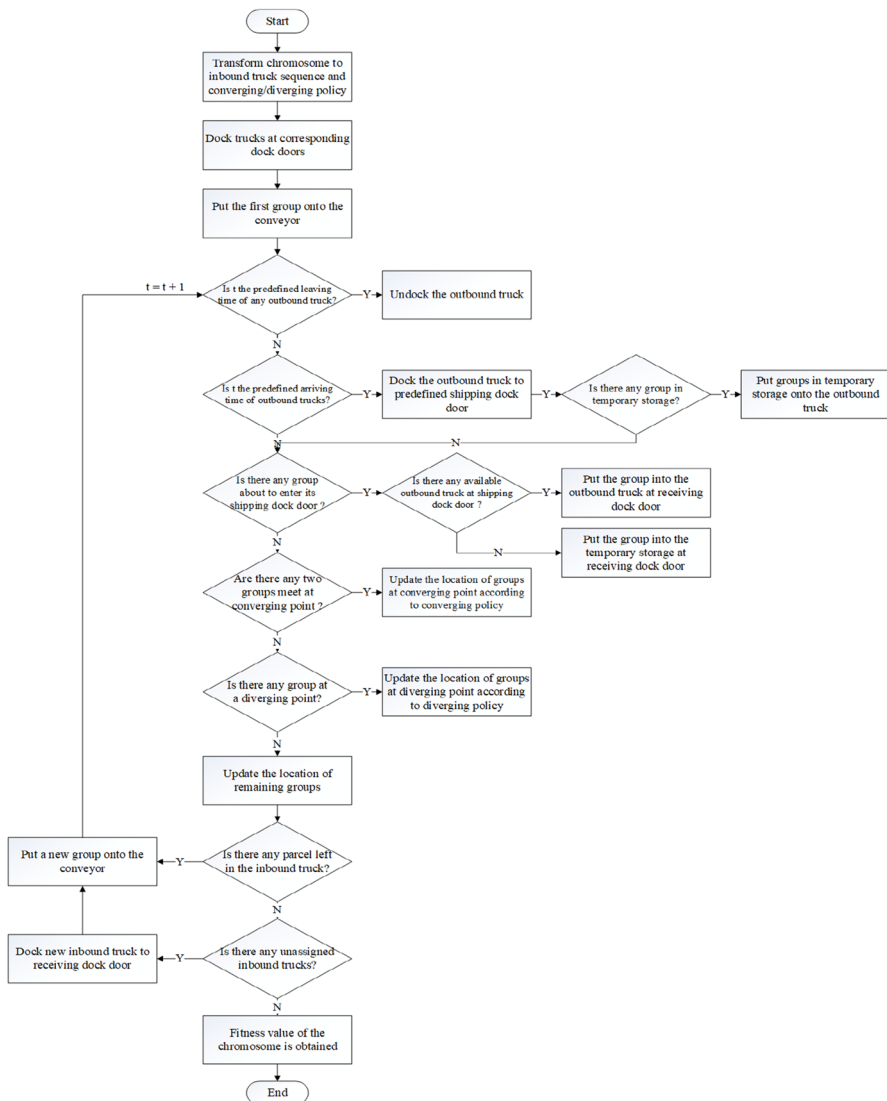
A1. Example of first and second sub-chromosome

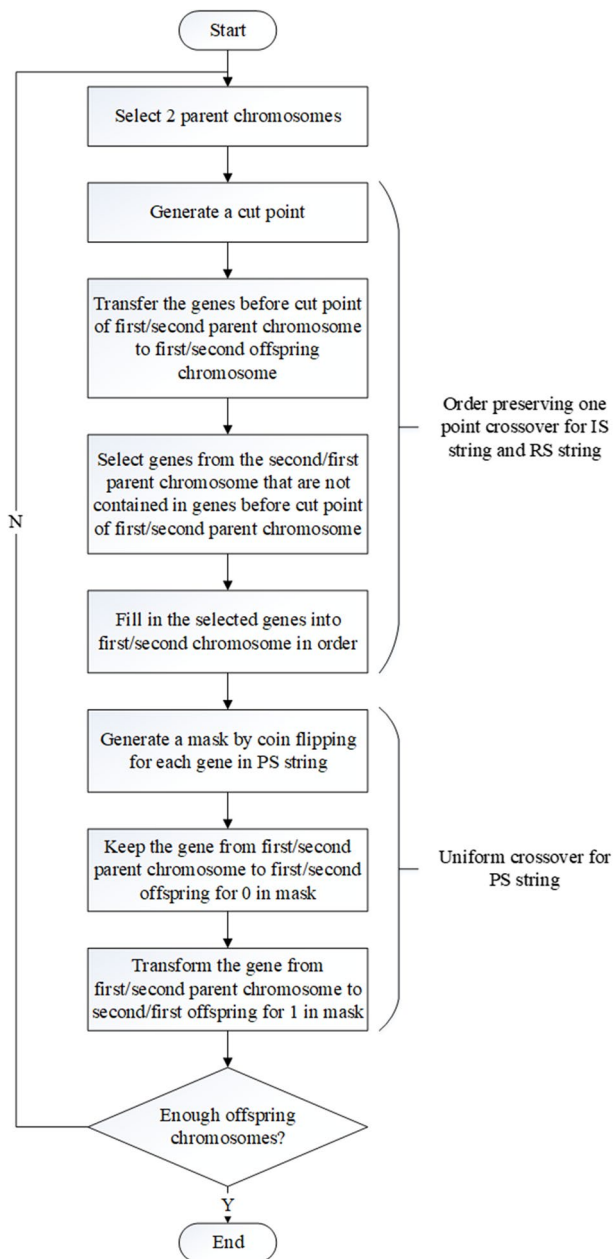


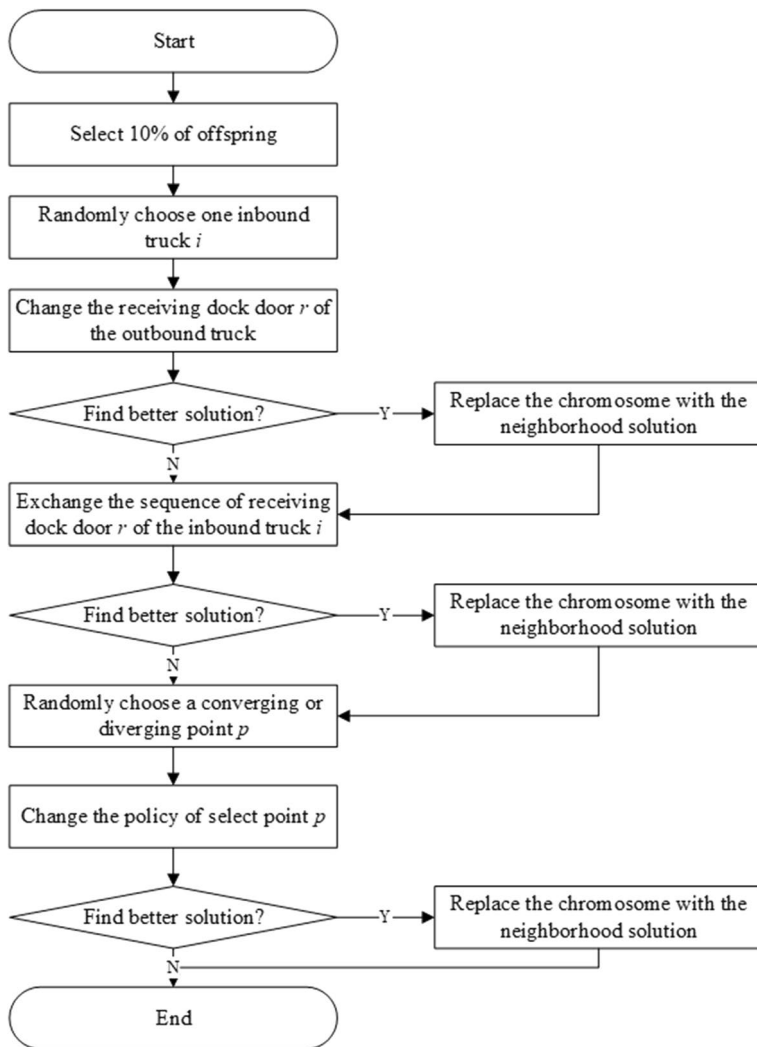
A2. Example of third sub-chromosome



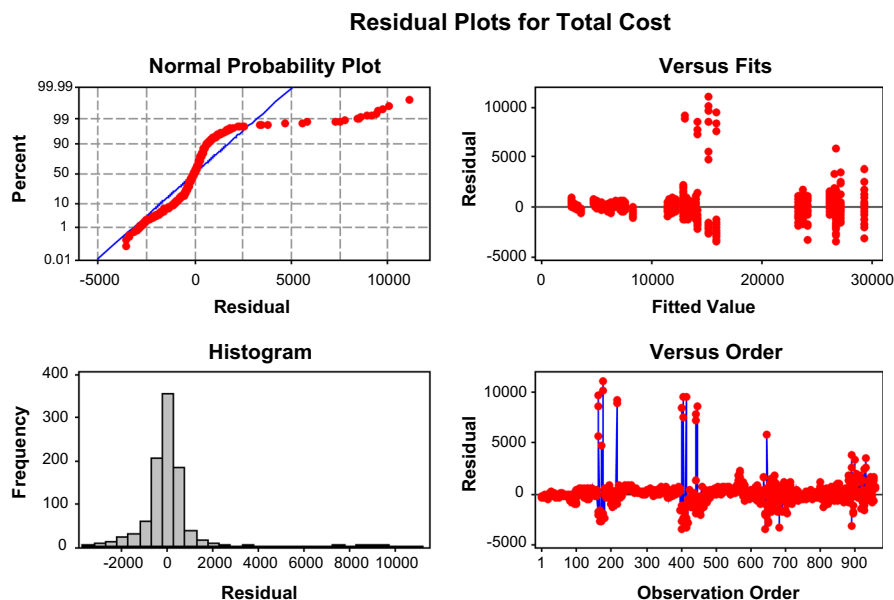
A3. Flowchart of fitness value evaluation



A4. Flow chart of crossover

A5. Flow chart of local search

A6. Residual plots for fitness value



References

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